GLOWINSKI'S METHOD FOR VARIATIONAL QUASILINEAR EQUATIONS WITH NONSTANDARD GROWTH: REGULARITY, DISCRETIZATION AND THE LAVRENTIEV PHENOMENON

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Abstract. The decomposition–coordination method of Glowinski and Marrocco, known today as ALG2 or the alternating direction method of multipliers, is a classical and very effective tool for the p-Laplacian. We show that it extends naturally to quasilinear equations with nonstandard growth, such as variable exponent, Orlicz, double phase and two-phase problems, and that its convergence is governed by the regularity of the minimizers. Beyond the convergence of the algorithm, we address a second and more delicate question: whether the computed solutions approximate the true minimizer. When a Lavrentiev gap is present, standard conforming finite elements converge to the wrong minimum. We show that, combined with nonconforming Crouzeix–Raviart elements, the method still works and, under suitable assumptions, yields computable two-sided bounds for the true minimum energy. Numerical experiments illustrate the theory.

1. Introduction

Let $\Omega \subset \mathbb{R}^d$ be a bounded open set and let $G\colon \Omega \times [0,\infty) \to [0,\infty)$ be such that $G(x,\cdot)$ is a convex Young function for a.e. x. We are interested in the numerical solution of the minimization problem

$$\min_v \ J(v), \qquad J(v) = \int_\Omega G(x,|\nabla v|)\,dx - \int_\Omega f v\,dx, \tag{1.1}$$

under Dirichlet boundary conditions. Its Euler–Lagrange equation is the *g-Laplace equation*

$$-\operatorname{div}\Big(g(x,|\nabla u|)\frac{\nabla u}{|\nabla u|}\Big) = f \quad \text{in } \Omega, \qquad g = \partial_t G. \tag{1.2}$$

This class is very large. Besides the p-Laplacian ($G = t^p/p$), it contains the $p(x)$-Laplacian of electrorheological fluids [2], double phase and two-phase energies $t^p/p + a(x)t^q/q$ with $a \geq 0$ [63, 26], Orlicz energies such as $t^p \log^a(1+t)$, and energies of exponential growth such as $e^t - 1 - t$, which violate the Δ_2 condition. Nonstandard growth brings two difficulties that are absent for the p-Laplacian with $p \geq 2$. The first is that the energy is no longer uniformly convex on L^2. The second is more serious: when G depends on x, smooth functions need not be dense in the energy space, and the infimum of J over smooth (or Lipschitz) functions may then be strictly larger than its infimum over all functions of finite energy. This is the Lavrentiev phenomenon [63, 64, 7], and it plays a central role in what follows.

A classical method. For the p-Laplacian, one of the most effective solvers is the decomposition–coordination method of Glowinski and Marrocco [45] and of Fortin and Glowinski [41, 42, 44], known as ALG2; with the step length $\rho = r$ used here, it is an alternating direction method of multipliers (ADMM) [46, 16]. The method introduces an auxiliary variable $\nu = \nabla u$ and a multiplier η for this constraint, and minimizes the corresponding augmented Lagrangian, with penalty parameter $r > 0$,

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alternately in u and in ν. Each iteration therefore consists of a *linear* Poisson-type problem for u, with the same matrix at every step; a *pointwise* nonlinear problem for ν, which reduces to a single scalar monotone equation $g(x,t) + rt = r|w|$ at each point; and an explicit update of η, which at convergence is the flux $g(x, |\nabla u|)\nabla u/|\nabla u|$. Since G enters only through the scalar equation, nothing in the algorithm depends on the growth of G. In particular, the code of [4] for the $p(x)$-Laplacian extends verbatim to arbitrary Young functions, and the conforming P1 discretization can be replaced by a Crouzeix–Raviart (CR) one without touching the nonlinear step.

Convergence is governed by regularity. Our first aim is to show that the theory can match this flexibility. Glowinski's convergence theorem requires the energy to be uniformly convex on bounded sets of L^2, and this fails as soon as $p < 2$ (Proposition 2.9). We show that uniform convexity is not needed. What governs the convergence of the continuous iteration is a regularity property of the solution, which we call Assumption (S): J has a minimizer in $H^1_0(\Omega)$ whose flux lies in $L^2(\Omega;\mathbb{R}^d)$, or, equivalently, the augmented Lagrangian has a saddle point. Under (S) the iterates converge weakly for every convex Young integrand, including nonsmooth integrands and integrands that violate Δ_2 (Theorem A). For strictly convex integrands they converge strongly, by the classical monotonicity argument, and for every Musielak–Orlicz integrand that satisfies Lieberman's condition $\delta \le tg'/g \le g_0$ uniformly in x, without any continuity in x, they converge in the natural energy space (Theorem B). This covers the p-Laplacian with $1 < p < 2$, the $p(x)$-Laplacian with a merely measurable exponent and $p^- < 2$, and two-phase energies with $p < 2$ and discontinuous coefficients, none of which is covered by Glowinski's theorem. Conversely, if (S) fails, the iterates are unbounded (Proposition 2.13, which follows from [37, Thm. 7]), so nothing weaker than (S) can suffice. Section 3 reviews when (S) is known to hold; for double phase, two-phase and exponential energies it is in general not available.

Does the computed solution approximate the true minimizer? Convergence of the algorithm is not the whole story. In practice the iteration is run on a finite element space, and there it always converges: for P1 and CR discretizations it converges for every penalty parameter $r > 0$ and every superlinear convex integrand, whatever the regularity of the solution (Theorem C). The real question is whether the discrete minimizers approximate the right object. If there is no Lavrentiev gap, conforming P1 elements are adequate. If there is one, the P1 energies converge to the infimum $m_{\mathcal{H}}$ of J over Lipschitz functions and not to the true infimum $m_{\mathcal{W}}$ over all functions of finite energy ([22]; see Proposition 5.1 for coefficients that are constant on the triangles). Conforming elements, in other words, converge to the wrong minimum. Balci, Ortner and Storn [8] showed that nonconforming CR elements do not suffer from this defect: under their assumptions, the CR energies converge to $m_{\mathcal{W}}$.

It is tempting to identify the failure of (S) with the presence of a gap, since both occur for the same kind of non-autonomous integrands. They are, however, different properties. Assumption (S) concerns the infimum over H^1_0, which lies between $m_{\mathcal{W}}$ and $m_{\mathcal{H}}$ (Proposition 3.7); it can fail without a gap (Example 3.9) and, at least in a degenerate example, hold together with one (Example 3.8).

Two-sided bounds for the minimum energy. Combined with CR elements, Glowinski's method thus converges on every mesh, and, under the assumptions of [8], its limits converge to the true minimum as the mesh is refined. Our main result makes this quantitative. Building on the discrete duality between CR and Raviart–Thomas elements [11], we show that every step of the CR iteration produces an equilibrated Raviart–Thomas field whose dual energy is a guaranteed lower bound for the *continuous* infimum $m_{\mathcal{W}}$, also in the presence of a Lavrentiev gap; when $f = 0$, the CR minimum itself is such a bound. This holds, essentially, when the mesh resolves the coefficients, the load is piecewise constant and the boundary datum is represented by its exact edge means; see Theorem D for the precise statement. Since every P1 energy is an upper bound for $m_{\mathcal{H}}$, one

obtains a computable bracket

$$D(\sigma) \le m_{\mathcal{W}} \le m_{\mathcal{H}} \le J(v_h),$$

whose width is a rigorous upper bound for the gap (Corollary 5.5). The bracket can certify the absence of a gap to any computed accuracy, but never its presence: no computable lower bound for $m_{\mathcal{H}}$ is available, so a gap can only be inferred from the behavior of the width under refinement. A second, qualitative diagnostic is the size of the discrete saddle points, which must blow up under refinement when (S) fails or when there is a gap (Theorem 5.7).

Numerical experiments. The experiments of Section 6 illustrate both halves of the theory. On smooth problems with exact solutions, the method converges with the expected orders for a wide range of integrands, including several that are not covered by Glowinski's theorem. On the standard Lavrentiev examples, namely Zhikov-type $p(x)$ and two-phase checkerboards and a double phase energy with the weight of Esposito, Leonetti and Mingione, we compute guaranteed brackets on meshes with up to 1.6×10^6 unknowns and compare them with the results of [8].

What is classical and what is new. The algorithm and its convergence theory under uniform convexity are due to Glowinski [44, Ch. VI], and so are several ingredients of our proofs: the Lyapunov inequality behind Theorem A is [44, Ch. VI, (5.25)], of which our identity (2.9) is the equality form for $\rho = r$; the characterization of saddle points and their independence of r (Proposition 2.3) is [44, Ch. VI, Thm. 2.1]; the weak convergence of the whole multiplier sequence by Opial's argument is [44, Ch. VI, §7]; and the existence of discrete saddle points is [44, Ch. VI, §4.2]. The convergence of ADMM in finite dimensions is [37, Thm. 8], and the convergence of CR energies to $m_{\mathcal{W}}$ is [8, Thm. 1]. The generalization of the Marini representation to nonlinear convex problems, that is, the equivalence of the CR method with a Raviart–Thomas mixed method with one-point quadrature and the absence of a discrete duality gap, is due to Carstensen and Liu [21, Thm. 3.1], for an autonomous integrand of quadratic growth arising in optimal design; see also [11] for general convex problems, [6, Prop. 2.37] for the $p(\cdot)$-Laplacian and, for the linear case, [53, 5]. Standard facts of convex analysis are collected, with references and without proofs, in Section 2.1, and results available in the literature are cited rather than proved. We give complete proofs wherever the setting (arbitrary Young integrands, nonsmooth energies, no growth conditions, CR spaces) or the exact hypotheses matter.

The remaining results are, as far as we know, new: the convergence theory under (S) alone (Theorems A and B), which covers exponents below 2, measurable coefficients and integrands without the Δ_2 condition; the account of when (S) holds and how it relates to the Lavrentiev gap (Section 3); the convergence of the discrete iteration for every r without any regularity (Theorem C); and, above all, the guaranteed lower bound for $m_{\mathcal{W}}$ at every CR iterate, the resulting bracket and the saddle-point diagnostic (Theorems D and 5.7, Corollary 5.5), together with the experiments that use them.

Limitations. Several questions remain open. Assumption (S) is a regularity property of the solution that the literature provides only in some cases; when it fails, the pair (η_n, ν_n) is unbounded, but the behavior of u_n is not known. We prove no rates, neither for the iteration nor for the discretization, so all orders reported in Section 6 are observations, and we treat only the step length $\rho = r$. The statement of Theorem D on the CR energy itself needs $f = 0$, and the theorem needs meshes that resolve the coefficients and exact edge means of the boundary datum. Finally, Sections 4–5 are written for $d = 2$.

Organization of the paper. Section 2 introduces the method in an abstract Hilbert space setting, explains what Glowinski's theorem covers, and proves Theorems A and B. Section 3 reviews when (S) is known to hold and separates it from the absence of a Lavrentiev gap. Section 4 introduces the P1 and CR discretizations and shows that the discrete iteration always converges. Section 5 turns to consistency: it shows that P1 energies are upper bounds for $m_{\mathcal{H}}$, proves the lower bound

of Theorem D and the bracket, and studies the size of discrete saddle points. Section 6 reports the experiments, and Section 7 collects conclusions and open problems.

Notation. We write $(\cdot,\cdot)$ and $\|\cdot\|$ for the inner product and the norm of $L^2(\Omega)$ or $L^2(\Omega;\mathbb{R}^d)$; other norms carry a subscript. The notation of convex analysis (domains, subdifferentials, conjugates) is recalled in Section 2.1. Weak convergence is denoted by $\rightharpoonup$. The boundary datum is always called ψ, and $m_{\mathcal{W}}$ and $m_{\mathcal{H}}$ denote the infima of J over finite-energy $W^{1,1}$ functions and over Lipschitz functions with boundary datum ψ (see Sections 3.3 and 5 for the precise definitions).

2. The method and its convergence: from the p-Laplacian to Musielak–Orlicz energies

This section develops the convergence theory of the method at the continuous level. We first recall the facts from convex analysis that we use (Section 2.1) and set up an abstract Hilbert space framework (Sections 2.2–2.4), then recall what Glowinski's classical theorem covers and why it stops at $p = 2$ (Section 2.5), and finally prove weak convergence under (S) for every convex Young integrand (Theorem A) and strong convergence for strictly convex integrands, in the energy space for Musielak–Orlicz integrands with Lieberman's structure (Theorem B).

2.1. **Preliminaries from convex analysis.** Let X be a real Hilbert space with inner product $(\cdot,\cdot)$. A function $\Psi\colon X\to(-\infty,+\infty]$ is *proper* if its domain $\operatorname{dom}\Psi=\{\Psi<\infty\}$ is nonempty, *convex* if $\Psi(sq+(1-s)q')\le s\Psi(q)+(1-s)\Psi(q')$ for $s\in[0,1]$, and *lower semicontinuous* (l.s.c.) if its sublevel sets are closed. Its *subdifferential* at q is $\partial\Psi(q)=\{\zeta\in X:\Psi(q')\ge\Psi(q)+(\zeta,q'-q)\ \forall q'\in X\}$, which is empty if $q\notin\operatorname{dom}\Psi$, and its *conjugate* is $\Psi^*(\zeta)=\sup_{q\in X}((\zeta,q)-\Psi(q))$. For a convex function γ on $[0,\infty)$, $\gamma'_\pm$ denote its one-sided derivatives and $\gamma^*(s)=\sup_{t\ge0}(st-\gamma(t))$, $s\ge0$, its (monotone) conjugate. We use the following standard facts.

- (P1) A convex l.s.c. function on X is weakly l.s.c. [38, Ch. I, Cor. 2.2], and a convex function that is bounded above on a neighborhood of a point is continuous at that point [38, Ch. I, §2].
- (P2) $\partial\Psi$ is monotone, $(\zeta-\zeta',q-q')\ge0$ for $\zeta\in\partial\Psi(q)$, $\zeta'\in\partial\Psi(q')$ (add the two subgradient inequalities); if Ψ is proper, convex and l.s.c., $\partial\Psi$ is maximal monotone [60].
- (P3) *Fenchel–Young.* $\Psi(q)+\Psi^*(\zeta)\ge(\zeta,q)$ whenever Ψ is proper, with equality if and only if $\zeta\in\partial\Psi(q)$ [38, Ch. I, Prop. 5.1], [59, Thm. 23.5].
- (P4) *Calculus.* q_0 minimizes Ψ if and only if $0\in\partial\Psi(q_0)$. If Ψ_1,Ψ_2 are convex and Ψ_1 is finite and continuous at a point of $\operatorname{dom}\Psi_2$, then $\partial(\Psi_1+\Psi_2)=\partial\Psi_1+\partial\Psi_2$ [38, Ch. I, Prop. 5.6]; in finite dimensions, if Ψ is finite and A is linear, $\partial(\Psi\circ A)=A^*\partial\Psi\circ A$ [59, Thm. 23.9]. In particular, if Ψ is proper, convex and l.s.c., $r>0$ and $w\in X$, then $\Psi+\frac{r}{2}\|\cdot-w\|^2$ has a unique minimizer p (the proximal point of Ψ/r at w), characterized by $r(w-p)\in\partial\Psi(p)$ [13, §12.4 and Ch. 16].
- (P5) *Radial functions.* If $\gamma\colon[0,\infty)\to[0,\infty)$ is finite, convex and nondecreasing with $\gamma(0)=0$, then $\phi(\xi)=\gamma(|\xi|)$ is convex on $\mathbb{R}^d$ and $\phi^*(\zeta)=\gamma^*(|\zeta|)$ [59, Thm. 15.3]; its subdifferential is described in Lemma 2.2 below.
- (P6) *Integral functionals.* Let $\Phi\colon\Omega\times\mathbb{R}^d\to[0,\infty)$ be measurable in x, convex and continuous in ξ, with $\Phi(x,0)=0$, and $I(q)=\int_\Omega\Phi(x,q)\,dx$ on $L^2(\Omega;\mathbb{R}^d)$. Then $I^*(\eta)=\int_\Omega\Phi^*(x,\eta)\,dx$ [58, Thm. 2], and hence, by (P3) applied to I and pointwise, $\eta\in\partial I(q)$ if and only if $q\in\operatorname{dom}I$ and $\eta(x)\in\partial\Phi(x,\cdot)(q(x))$ for a.e. x.

Definition 2.1 (Young integrand)**.** A function $G\colon\Omega\times[0,\infty)\to[0,\infty)$ is a *Young integrand* if $G(\cdot,t)$ is measurable for every $t\ge0$ and, for a.e. $x\in\Omega$, $G(x,\cdot)$ is finite, convex and nondecreasing with $G(x,0)=0$.

A finite convex nondecreasing function on $[0,\infty)$ that vanishes at 0 is continuous, so G is a Carathéodory function. We write

$$\Phi(x,\xi) = G(x,|\xi|), \qquad g(x,t) = \partial_t^+ G(x,t) \quad (t \geq 0).$$

Then $\Phi(x,\cdot)$ is convex on $\mathbb{R}^d$ by (P5), and $g(x,\cdot)$ is nondecreasing, finite and nonnegative, with $g(x,0) = \inf_{s>0} G(x,s)/s$. The integrand is *superlinear* if $G(x,t)/t \to \infty$ as $t \to \infty$, equivalently $g(x,t) \to \infty$; it satisfies Δ_2 if $G(x,2t) \leq KG(x,t)$; and it satisfies Lieberman's condition [50, (1.1′)] if $\delta \leq tg'/g \leq g_0$, see (L) below, which implies Δ_2 (see (2.12)). No growth condition (Δ_2, ∇_2 or superlinearity) is assumed in this section. The subdifferential of a radial function is elementary.

Lemma 2.2 (Subdifferential of Φ)**.** *Let $\gamma\colon [0,\infty) \to [0,\infty)$ be finite, convex, nondecreasing, $\gamma(0) = 0$, and $\phi(\xi) = \gamma(|\xi|)$.* (i) *$\partial\phi(0) = \{\zeta : |\zeta| \leq \gamma'_+(0)\}$.* (ii) *If $\xi \neq 0$ and γ is differentiable at $|\xi|$, then $\partial\phi(\xi) = \{\gamma'(|\xi|)\xi/|\xi|\}$.* (iii) *If $\gamma \in C^1([0,\infty))$ with $\gamma'(0) = 0$, then $\phi \in C^1(\mathbb{R}^d)$ and $\partial\phi(\xi) = \{T(\xi)\}$, $T(\xi) := \gamma'(|\xi|)\xi/|\xi|$, $T(0) = 0$.*

Proof. (i) $\zeta \in \partial\phi(0)$ iff $\gamma(t) \geq |\zeta|t$ for all $t \geq 0$, iff $|\zeta| \leq \inf_{t>0} \gamma(t)/t = \gamma'_+(0)$. (ii)–(iii) A convex function differentiable at ξ has $\partial\phi(\xi) = \{\nabla\phi(\xi)\}$ [59, Thm. 25.1]; in (iii), $\phi(\xi) = o(|\xi|)$ at 0. □

2.2. **Setting.** Let $V = H_0^1(\Omega)$ with $\|v\|_V = \|\nabla v\|$, $H = L^2(\Omega;\mathbb{R}^d)$, $f \in L^2(\Omega)$, and

$$F(q) = \int_\Omega \Phi(x,q(x))\,dx \in [0,+\infty] \quad (q \in H), \qquad J(v) = F(\nabla v) - (f,v) \quad (v \in V).$$

The functional F is convex, satisfies $F \geq 0 = F(0)$, and is lower semicontinuous on H by Fatou's lemma applied along a.e. convergent subsequences; hence it is also weakly lower semicontinuous by (P1).

Abstract structure. We write $B = \nabla\colon V \to H$ and $\ell(v) = (f,v)$. Sections 2.3–2.6 use only the following properties:

(A1) V and H are real Hilbert spaces and $B\colon V \to H$ is linear with $\|Bv\|_H = \|v\|_V$;
(A2) $F\colon H \to [0,+\infty]$ is convex, lower semicontinuous and proper;
(A3) $\ell \in V^*$.

Working at this level of generality costs nothing, and it allows us to apply the same proofs verbatim to the finite element problems of Section 4.

2.3. **Augmented Lagrangian and saddle points.** The method is built on the augmented Lagrangian associated with the constraint $q = Bv$. For $r > 0$ we define on $V \times H \times H$

$$\mathcal{L}_r(v,q,\mu) = F(q) - \ell(v) + (\mu, Bv - q) + \frac{r}{2}\|Bv - q\|^2.$$

A triple (u^*,ν^*,η^*) is a saddle point if $\mathcal{L}_r(u^*,\nu^*,\mu) \leq \mathcal{L}_r(u^*,\nu^*,\eta^*) \leq \mathcal{L}_r(v,q,\eta^*)$ for all (v,q,μ), with the middle term finite. The next proposition characterizes saddle points by the optimality system of the problem.

Proposition 2.3 (Saddle points [44, Ch. VI, Thm. 2.1])**.** *Assume (A1)–(A3). (u^*,ν^*,η^*) is a saddle point of $\mathcal{L}_r$ if and only if*

$$\nu^* = Bu^*, \qquad \eta^* \in \partial F(\nu^*), \qquad (\eta^*, Bw) = \ell(w) \quad \forall w \in V. \tag{2.1}$$

In that case $F(\nu^*) < \infty$ and u^* minimizes $J(v) = F(Bv) - \ell(v)$ over V. The set of saddle points does not depend on r.

Glowinski treats $\min_v F(Bv) + \mathcal{G}(v)$ with a convex, proper, l.s.c. $\mathcal{G}$; for $\mathcal{G} = -\ell$, his characterization [44, Ch. VI, (2.11)–(2.12)] of saddle points (for which $F(\nu^*) - \ell(u^*)$ is finite) reads $\eta^* \in \partial F(\nu^*)$ and $(\eta^*, B(v-u^*)) \geq \ell(v-u^*)$ for all $v \in V$, which together with $\nu^* = Bu^*$ is (2.1).

Assumption S. The augmented Lagrangian $\mathcal{L}_r$ has a saddle point in $V \times H \times H$; equivalently, (2.1) has a solution.

In the concrete case, (S) says that J has a minimizer $u \in H_0^1(\Omega)$ whose *flux* $\eta \in \partial F(\nabla u)$ belongs to $L^2(\Omega;\mathbb{R}^d)$ and satisfies $(\eta, \nabla w) = (f, w)$ for all $w \in H_0^1$. It is therefore a property of the solution, and not only of G; Section 3 discusses when it holds. To work with (S) we need a pointwise description of ∂F, which the next lemma provides together with Lemma 2.2.

Lemma 2.4 (Pointwise description of ∂F)**.** *Let $q \in \operatorname{dom} F$ and $\eta \in H$. Then $\eta \in \partial F(q)$ if and only if $\eta(x) \in \partial\Phi(x,\cdot)(q(x))$ for a.e. $x \in \Omega$.*

Proof. This is (P6) with $I = F$, Φ being a Carathéodory integrand. □

Consequently, if $G(x,\cdot) \in C^1([0,\infty))$ with $g(x,0) = 0$, then for $q \in \operatorname{dom} F$

$$\partial F(q) = \{T(\cdot,q)\} \ \text{ if } T(\cdot,q) \in H, \qquad \partial F(q) = \emptyset \ \text{ otherwise;} \qquad T(x,\xi) = g(x,|\xi|)\frac{\xi}{|\xi|}. \tag{2.2}$$

For $G = e^t - 1$, by contrast, $g(x,0) = 1$, and ∂F is multivalued at every field that vanishes on a set of positive measure.

2.4. **The algorithm.** Fix $r > 0$, $\nu_0 \in H$ and $\eta_1 \in H$ (our code uses $\nu_0 = \eta_1 = 0$). For $n \geq 1$ the iteration reads

$$u_n \in V: \quad (\eta_n + r(Bu_n - \nu_{n-1}), Bw) = \ell(w) \quad \forall w \in V, \tag{2.3}$$

$$\nu_n = \operatorname*{argmin}_{q\in H} \Big\{ F(q) - (\eta_n, q) + \frac{r}{2}\|Bu_n - q\|^2 \Big\}, \tag{2.4}$$

$$\eta_{n+1} = \eta_n + r(Bu_n - \nu_n). \tag{2.5}$$

Step (2.3) minimizes $\mathcal{L}_r(\cdot, \nu_{n-1}, \eta_n)$, step (2.4) minimizes $\mathcal{L}_r(u_n, \cdot, \eta_n)$, and (2.5) is a multiplier step of length $\rho = r$. This is ALG2 of [42], [44, Ch. VI, (3.7)–(3.10)] with $\rho = r$ and the same order of the steps, as used in [4]. In the concrete case, (2.3) is the Poisson problem $-r\Delta u_n = f + \operatorname{div}(\eta_n - r\nu_{n-1})$, $u_n \in H_0^1$. We call $s_n = Bu_n - \nu_n$ the primal residual, so that $\eta_{n+1} - \eta_n = r s_n$. The following lemmas show that every step is well defined and that the nonlinear step decouples into scalar problems.

Lemma 2.5 (Well-posedness)**.** *Assume (A1)–(A3).* (i) (2.3) *has a unique solution.* (ii) (2.4) *has a unique solution ν, characterized by $\eta_n + r(Bu_n - \nu) \in \partial F(\nu)$; consequently*

$$\eta_{n+1} \in \partial F(\nu_n) \qquad (n \geq 1). \tag{2.6}$$

(iii) *For $n \geq 1$,*

$$(\eta_{n+1} + r(\nu_n - \nu_{n-1}), Bw) = \ell(w) \qquad \forall w \in V. \tag{2.7}$$

Proof. (i) This is Riesz' theorem in V with the inner product $r(Bv, Bw)$. (ii) Up to a constant, the functional in (2.4) is $F(q) + \frac{r}{2}\|q - w\|^2$ with $w = Bu_n + \eta_n/r$, so (P4) gives a unique minimizer ν, characterized by $r(w - \nu) = \eta_n + r(Bu_n - \nu) \in \partial F(\nu)$. (iii) It suffices to insert $\eta_n + r(Bu_n - \nu_{n-1}) = \eta_{n+1} + r(\nu_n - \nu_{n-1})$ into (2.3). □

Lemma 2.6 (The pointwise problem)**.** *Let γ and ϕ be as in Lemma 2.2, $r > 0$, $w \in \mathbb{R}^d$.*

(i) *$\xi \mapsto \phi(\xi) + \frac{r}{2}|\xi - w|^2$ has a unique minimizer $\hat\xi = \hat t\, w/|w|$ ($\hat\xi = 0$ if $w = 0$), where $\hat t \in [0, |w|]$ minimizes $\chi(t) = \gamma(t) + \frac{r}{2}(t - |w|)^2$ on $[0,\infty)$.*

(ii) *$\hat t = 0$ if and only if $r|w| \leq \gamma'_+(0)$; otherwise $\gamma'_-(\hat t) \leq r(|w| - \hat t) \leq \gamma'_+(\hat t)$. If $\gamma \in C^1$ and $\gamma'(0) < r|w|$, $\hat t$ is the unique root in $(0, |w|]$ of the increasing function $t \mapsto \gamma'(t) + rt - r|w|$.*

(iii) *$\hat\eta := r(w - \hat\xi) \in \partial\phi(\hat\xi)$ and $|\hat\eta| = r(|w| - \hat t)$.*

Proof. (i) This is the formula for the proximal point of a function of the norm [13, Ch. 24], applied to the even extension of γ, which is convex since γ is nondecreasing. Directly: χ is strictly convex and coercive, $\chi(t) > \chi(|w|)$ for $t > |w|$, and $\phi(\xi) + \frac{r}{2}|\xi - w|^2 \geq \chi(|\xi|) \geq \chi(\hat t)$ because $|\xi - w| \geq ||\xi| - |w||$, with equality at $\hat\xi$. (ii) This is the optimality condition $\chi'_+(\hat t) \geq 0 \geq \chi'_-(\hat t)$ for the one-dimensional problem. (iii) This is the characterization of the minimizer in (P4). □

Corollary 2.7 (Pointwise form of (2.4))**.** *In the concrete case, let $w_n = \eta_n/r + \nabla u_n$. For a.e. x, $\nu_n(x) = t(x)\, w_n(x)/|w_n(x)|$, where $t(x) = 0$ if $r|w_n(x)| \le g(x,0)$ and otherwise, when $G(x,\cdot) \in C^1$, $t(x) \in (0, |w_n(x)|]$ solves*

$$g(x,t) + r\,t = r|w_n(x)|. \tag{2.8}$$

Moreover $\eta_{n+1}(x) = r(w_n(x) - \nu_n(x)) \in \partial\Phi(x,\cdot)(\nu_n(x))$: the multiplier is the flux.

Proof. By (2.6) and Lemma 2.4, $\eta_{n+1}(x) = r(w_n(x) - \nu_n(x)) \in \partial\Phi(x,\cdot)(\nu_n(x))$ a.e., so $\nu_n(x)$ minimizes $\Phi(x,\cdot) + \frac{r}{2}|\cdot - w_n(x)|^2$; apply Lemma 2.6. □

2.5. **What Glowinski's theorem covers: from $p \ge 2$ to nonstandard growth.** Before developing our own theory, we recall the classical convergence result and determine its scope on $H = L^2$. Glowinski [44, Ch. VI, §§4–5] considers $\min_{v\in V} F(Bv) + \Gamma(v)$ with F, Γ proper, convex and l.s.c., and assumes

(i) the augmented Lagrangian has a saddle point;
(ii) B is injective and $B(V)$ is closed in H;
(iii) $F(q)/\|q\| \to +\infty$ as $\|q\| \to \infty$;
(iv) $F = F_0 + F_1$ with F_0, F_1 proper, convex, l.s.c., F_0 Gâteaux differentiable and uniformly convex on bounded sets in the sense of [44, (4.6)]: for every M there is a continuous strictly increasing δ_M, $\delta_M(0) = 0$, with $(F_0'(q) - F_0'(p), q - p) \ge \delta_M(\|q - p\|)$ for $\|p\|, \|q\| \le M$.

Under these hypotheses, [44, Thm. 5.1] asserts that ALG2 with $0 < \rho < \frac{1+\sqrt{5}}{2} r$ satisfies $u_n \to u$ strongly in V and $\nu_n \to Bu$ strongly in H, and that (η_n) is bounded, with multipliers as weak cluster points; by [44, §7], the whole sequence (η_n) converges weakly. For linear Γ, as here, the range $0 < \rho < 2r$ is recorded in [44, Rem. 5.1, §6.3]. Glowinski himself treats the p-Laplacian in its natural setting $V = W_0^{1,p}(\Omega)$, $H = L^p(\Omega)^d$ [44, §6.3]; since these spaces are not Hilbert spaces for $p \ne 2$, he observes that his convergence theorems do not apply to the continuous problem, but only to its finite element approximations, which are finite dimensional. We proceed differently: we keep $V = H_0^1(\Omega)$ and $H = L^2(\Omega)^d$ for every integrand, which is what the algorithm actually computes (the u-step is a Poisson problem), and pay for this with Assumption (S). On $V = H_0^1$ and $H = L^2$, hypothesis (ii) is trivial, and (iii)–(iv) hold for a large class of integrands.

Lemma 2.8 (Glowinski's hypotheses on $H = L^2$)**.** *Let G be a Young integrand.*

(a) *If $t \mapsto G(x,t) - \frac{\varepsilon}{2}t^2$ is convex for some $\varepsilon > 0$ and a.e. x, then $F_0 = \frac{\varepsilon}{2}\|\cdot\|^2$, $F_1 = F - F_0$ satisfy* (iii)–(iv).
(b) *Let $G(x,\cdot) \in C^1$, $g(x,0) = 0$, $k(x,t) = g(x,t)/t$ nondecreasing in t, $k_*(t) := \operatorname{ess\,inf}_x k(x,t) > 0$ for $t > 0$ and $K := \operatorname{ess\,sup}_x k(x,1) < \infty$. Let $h(x,t) = G(x,t)$ for $t \le 1$ and $h(x,t) = G(x,1) + \frac{1}{2}k(x,1)(t^2 - 1)$ for $t > 1$. Then $F_0 = \int_\Omega h(x,|q|)\,dx$ and $F_1 = F - F_0$ satisfy* (iii)–(iv).

Proof. (a) $G(x,\cdot) - \frac{\varepsilon}{2}t^2$ is convex with nonnegative right derivative at 0, hence nondecreasing, so F_1 is convex, proper and l.s.c.; $F \ge F_0$ gives (iii). (b) $\partial_t h(x,t) = t\,k(x, \min\{t,1\}) \le Kt$, so F_0 is C^1 on H with Lipschitz gradient $q \mapsto k(\cdot, \min\{|q|,1\})q$; $G - h$ is convex and nondecreasing because $g(x,t) - g(x,s) \ge (t-s)k(x,1)$ for $t > s \ge 1$. Lemma 2.14 below, applied to $k(x, \min\{t,1\})$, and the splitting argument of the proof of Theorem B(b) give $(F_0'(q) - F_0'(p), q - p) \ge \frac{3}{8}k_*(\min\{\frac{\|q-p\|}{4|\Omega|^{1/2}}, 1\})\|q - p\|^2$, and replacing k_* by the continuous minorant $\tau^{-1}\int_0^\tau k_*$ yields an admissible δ_M. Finally $G \ge h \ge \frac{1}{2}k_*(1)(t^2 - 1)$ gives (iii). □

Part (b) of Lemma 2.8 covers t^p/p and $t^{p(x)}/p(x)$ with $2 \le p(x) \le p^+ < \infty$, which justifies [4, Thm. 4.3]; part (a) covers uniformly convex integrands such as $e^t - 1$, $e^t - 1 - t$, $t^2/2 + a(x)t^4/4$ with measurable $a \ge 0$, and $t^{3/2}/(3/2) + t^3/3$. For all of these, Glowinski's theorem gives strong convergence under (S). Below $p = 2$, however, it gives nothing.

Proposition 2.9. *Let $F(q) = \frac{1}{p}\int_\Omega |q|^p\,dx$ on $H = L^2(\Omega;\mathbb{R}^d)$ with $1 < p < 2$. Then F satisfies neither* (iii) *nor* (iv).

Proof. Fix a unit vector e and, for large M, $A_M \subset \Omega$ with $|A_M| = M^{-2+\varepsilon}$, $0 < \varepsilon < 2(2-p)$; $q_M = M\mathbf{1}_{A_M}e$ has $\|q_M\| = M^{\varepsilon/2} \to \infty$ and $F(q_M)/\|q_M\| = M^{p-2+\varepsilon/2}/p \to 0$: (iii) fails. With $|A_M| = M^{-2}$ instead, $\|q_M\| = 1$ and $\frac{1}{2}(F(0)+F(q_M)) - F(\frac{q_M}{2}) = \frac{M^{p-2}}{p}(\frac{1}{2} - 2^{-p}) \to 0$. If $F = F_0 + F_1$ as in (iv), then with $m = q_M/2$ and $q \in \{0, q_M\}$, $F_0(q) - F_0(m) - (F_0'(m), q-m) = \int_0^1 (F_0'(m+s(q-m)) - F_0'(m), q-m)\,ds \ge \int_0^1 \delta_1(s/2)\frac{ds}{s} \ge \frac{1}{2}\delta_1(\frac{1}{4})$; averaging over q, $\frac{1}{2}(F_0(0) + F_0(q_M)) - F_0(m) \ge \frac{1}{2}\delta_1(\frac{1}{4}) > 0$, while the same expression for F_1 is ≥ 0. This contradicts the previous computation. □

The proof localizes: for $G = t^{p(x)}/p(x)$ with $|\{p \le p_0\}| > 0$ for some $p_0 < 2$, choosing $A_M \subset \{p \le p_0\}$ gives the same conclusions. Thus the classical theorem covers neither the p-Laplacian with $p < 2$ nor variable exponents with $p^- < 2$. Nothing in the *algorithm*, on the other hand, requires Δ_2 or ∇_2: the steps (2.3)–(2.5) are defined for every Young integrand. The next two subsections develop a theory that matches this generality. We prove weak convergence under (S) alone, for every convex Young integrand (Theorem A), and strong convergence for strictly convex integrands, in the energy space for every Musielak–Orlicz integrand with Lieberman's structure, uniformly in x but without any continuity in x (Theorem B).

2.6. Theorem A: weak convergence under (S).

Remark 2.10 (Uniqueness of saddle points). By Proposition 2.3, every saddle point has the form $(u^*, \nabla u^*, \eta^*)$, where u^* minimizes J and $\eta^* \in \partial F(\nabla u^*)$ satisfies $(\eta^*, \nabla w) = \ell(w)$ for all w. Two situations are therefore relevant. If J has a unique minimizer, for instance if $G(x,\cdot)$ is strictly convex, all saddle points share the first two components u^* and $\nu^* = \nabla u^*$. If, in addition, ∂F is single-valued at ν^*, which holds when $G(x,\cdot) \in C^1$ with $g(x,0) = 0$ (Lemma 2.2), then η^* is the flux $g(x,|\nabla u^*|)\nabla u^*/|\nabla u^*|$ and the saddle point, if it exists, is unique. What matters is the single-valuedness of ∂F at ν^*, not that F be C^1 on H, which fails for $p > 2$ since $F = +\infty$ outside L^p. Uniform convexity alone does not suffice: $G = e^t - 1$ is uniformly convex, but its multiplier is not determined in the dead core, where $\nabla u^* = 0$ and ∂F is multivalued; and for integrands with affine pieces the minimizer itself need not be unique [21].

The first result shows that (S) alone yields a Lyapunov functional, and with it weak convergence of the whole sequence of iterates.

Theorem A. *Assume (A1)–(A3) and Assumption S. Let (u_n, ν_n, η_n) be generated by (2.3)–(2.5), let (u^*, ν^*, η^*) be any saddle point, and set*

$$E_n = \frac{1}{2r}\|\eta_{n+1} - \eta^*\|^2 + \frac{r}{2}\|\nu_n - \nu^*\|^2 \qquad (n \ge 0).$$

Then for every $n \ge 1$

$$E_n + \frac{r}{2}\|s_n\|^2 + \frac{r}{2}\|\nu_n - \nu_{n-1}\|^2 + (\eta_{n+1} - \eta^*, \nu_n - \nu^*) + r(s_n, \nu_n - \nu_{n-1}) = E_{n-1}, \tag{2.9}$$

and the last two terms on the left are nonnegative (the last one for $n \ge 2$, and for $n = 1$ if $\eta_1 \in \partial F(\nu_0)$). Consequently (E_n) is nonincreasing; (η_n), (ν_n) are bounded in H and (u_n) in V;

$$\sum_{n\ge1}\|s_n\|^2 < \infty, \qquad \sum_{n\ge1}\|\nu_n - \nu_{n-1}\|^2 < \infty, \qquad \sum_{n\ge1} m_n < \infty, \quad m_n := (\eta_{n+1} - \eta^*, \nu_n - \nu^*) \ge 0.$$

Moreover there is a saddle point $(\bar u, \bar\nu, \bar\eta)$ with $u_n \rightharpoonup \bar u$ in V, $\nu_n \rightharpoonup \bar\nu$ and $\eta_n \rightharpoonup \bar\eta$ in H. In the concrete case $u_n \to \bar u$ strongly in $L^2(\Omega)$, $\nabla u_n - \nu_n \to 0$ in L^2, and $\bar u$ minimizes J over $H_0^1(\Omega)$.

Neither a growth condition nor the differentiability of G is used. Whenever (S) holds, Theorem A therefore applies to t^p/p for every $p > 1$, to variable exponent, double phase and two-phase energies, to $t^p\log^a(1+t)$, and to $e^t - 1 - t$ and $e^t - 1$.

Proof. Step 1: the identity. We subtract $(\eta^*, Bw) = \ell(w)$ from (2.7) and take $w = u_n - u^*$, so that $Bw = s_n + (\nu_n - \nu^*)$:

$$(\eta_{n+1} - \eta^*, s_n) + (\eta_{n+1} - \eta^*, \nu_n - \nu^*) + r(\nu_n - \nu_{n-1}, s_n) + r(\nu_n - \nu_{n-1}, \nu_n - \nu^*) = 0.$$

By $2(a-c, a-b) = \|a-c\|^2 + \|a-b\|^2 - \|b-c\|^2$ and $rs_n = \eta_{n+1} - \eta_n$, the first term equals $\frac{1}{2r}(\|\eta_{n+1} - \eta^*\|^2 - \|\eta_n - \eta^*\|^2) + \frac{r}{2}\|s_n\|^2$ and the last one equals $\frac{r}{2}(\|\nu_n - \nu^*\|^2 - \|\nu_{n-1} - \nu^*\|^2) + \frac{r}{2}\|\nu_n - \nu_{n-1}\|^2$. This is (2.9).

Step 2: signs. By (2.6), (2.1) and the monotonicity of ∂F (P2), $m_n \geq 0$. Similarly, $\eta_n \in \partial F(\nu_{n-1})$ gives $r(s_n, \nu_n - \nu_{n-1}) = (\eta_{n+1} - \eta_n, \nu_n - \nu_{n-1}) \geq 0$. Summing (2.9) yields the bounds and the summability, and $Bu_n = s_n + \nu_n$ is bounded.

Step 3: weak cluster points are saddle points. Let $\nu_{n_k} \rightharpoonup \bar{\nu}$, $\eta_{n_k+1} \rightharpoonup \bar{\eta}$ and $u_{n_k} \rightharpoonup \bar{u}$. Since $s_n \to 0$, we have $B\bar{u} = \bar{\nu}$, and passing to the limit in (2.7) gives $(\bar{\eta}, Bw) = \ell(w)$. Taking $w = u_n$ in (2.7), $(\eta_{n+1}, \nu_n) = \ell(u_n) - (\eta_{n+1}, s_n) - r(\nu_n - \nu_{n-1}, Bu_n) \to \ell(\bar{u}) = (\bar{\eta}, \bar{\nu})$ along n_k. By (2.6), $F(q) \geq F(\nu_{n_k}) + (\eta_{n_k+1}, q) - (\eta_{n_k+1}, \nu_{n_k})$; by weak lower semicontinuity (P1), $F(q) \geq F(\bar{\nu}) + (\bar{\eta}, q - \bar{\nu})$. Hence $\bar{\eta} \in \partial F(\bar{\nu})$, and $(\bar{u}, \bar{\nu}, \bar{\eta})$ is a saddle point by Proposition 2.3.

Step 4: the whole sequence. Let $x_n = (\nu_n, \eta_{n+1})$ in $H \times H$ with the inner product $r(\nu, \nu') + \frac{1}{r}(\eta, \eta')$. For every saddle pair $z = (\nu^*, \eta^*)$, $|x_n - z|^2 = 2E_n(z)$ converges by Step 2, and every weak cluster point of (x_n) is a saddle pair by Step 3; by Opial's lemma [56], as in [44, Ch. VI, §7], $x_n \rightharpoonup (\bar{\nu}, \bar{\eta})$. Hence $\eta_n \rightharpoonup \bar{\eta}$, and, since B is injective, $u_n \rightharpoonup \bar{u}$. In the concrete case, Rellich's theorem gives $u_n \to \bar{u}$ in L^2. □

Remark 2.11 (Identification of the limit). In the situation of Remark 2.10, the limit in Theorem A is the saddle point used to define E_n: if J has a unique minimizer, then $\bar{u} = u^*$ and $\bar{\nu} = \nu^*$, and if moreover ∂F is single-valued at ν^*, then $(\bar{u}, \bar{\nu}, \bar{\eta}) = (u^*, \nu^*, \eta^*)$ and the whole sequence converges weakly to the unique saddle point, whatever the initialization. Without uniqueness this need not be so. The functional E_n is nonincreasing for *every* saddle point but need not tend to zero, and the limit may depend on the initialization: if $\nu_0 = \bar{\nu}$ and $\eta_1 = \bar{\eta}$ for some saddle point $(\bar{u}, \bar{\nu}, \bar{\eta})$, the iterates stay there, and E_n is the constant distance from it to (u^*, ν^*, η^*).

Remark 2.12 (Initialization; related results). Since $s_n + (\nu_n - \nu_{n-1}) = Bu_n - \nu_{n-1}$, identity (2.9) can be written as $E_n + m_n + \frac{r}{2}\|Bu_n - \nu_{n-1}\|^2 = E_{n-1}$ for every $n \geq 1$, whatever η_1 is; hence all conclusions hold for arbitrary $\nu_0, \eta_1 \in H$. Identity (2.9) is the equality version, for $\rho = r$, of Glowinski's inequality [44, Ch. VI, (5.25)], and Step 4 is the argument of [44, Ch. VI, §7]; what Theorem A adds is that hypotheses (iii)–(iv) are not needed for weak convergence. The same functional appears in [12, Thm. 3.7] and [16, App. A]. Weak convergence can also be derived from the equivalence of ADMM with Douglas–Rachford splitting for the dual problem [43], [37, §5] and from the weak convergence of the latter [51, 61, 13]; see [54]. This weak convergence cannot be improved in general. Indeed, through the correspondence in the proof of Proposition 2.13, the example of Bùi and Combettes [19], in which the Douglas–Rachford shadow sequence converges weakly but not strongly, is an instance of (A1)–(A3) ($V = \mathbb{R}$, $Bt = te_0$, $\ell = 0$) in which that sequence is η_n. The example says nothing, however, about ν_n or u_n.

The Lyapunov identity also shows that (S) cannot be dispensed with: without a saddle point, the iterates cannot remain bounded.

Proposition 2.13 ((S) is necessary for boundedness)**.** *Assume (A1)–(A3) and let (u_n, ν_n, η_n) be generated from any $\nu_0, \eta_1 \in H$. Then* (S) *holds if and only if $(\eta_n + r\nu_{n-1})_n$ is bounded in H. In particular, if* (S) *fails, (η_n) or (ν_n) is unbounded.*

Proof. If (S) holds, boundedness follows from Theorem A and Remark 2.12. Conversely, (2.3)–(2.5) is the ADMM of [37, Thm. 8] for $\min(-\ell(v)) + F(Bv)$ with penalty r, relaxation 1 and exact steps; the relations (Y1)–(Y3) of its proof are algebraic and hold in Hilbert space: $z_n := \eta_n + r\nu_{n-1}$ is the Douglas–Rachford sequence for the maximal monotone (P2) operators $\mathcal{A} = \partial\iota_{\{B^*\eta = \ell\}}$ and $\mathcal{B} = \partial F^*$.

The affine space $\{B^*\eta = \ell\}$ is nonempty and closed (B^* is onto by (A1)) with normal cone $R(B)$, and $0 \in \mathcal{A}\eta + \mathcal{B}\eta$ means $\eta \in \partial F(Bu)$ for some u with $B^*\eta = \ell$, i.e. (2.1). If (S) fails, $\operatorname{zer}(\mathcal{A}+\mathcal{B}) = \emptyset$ and (z_n) is unbounded by [37, Thm. 7], stated in a general Hilbert space. □

2.7. **Theorem B: strong convergence for Musielak–Orlicz integrands.** Weak convergence is the best one can expect in the abstract setting, but for concrete integrands the monotonicity of the flux gives more. Assume now that $G(x,\cdot) \in C^1([0,\infty))$ with $g(x,0) = 0$ for a.e. x, and let $T(x,\xi) = g(x,|\xi|)\xi/|\xi|$,

$$k(x,t) = \frac{g(x,t)}{t}\ (t>0), \qquad \varpi(x;a,b) = (T(x,a) - T(x,b))\cdot(a-b) \geq 0.$$

By (2.2) and (2.6), $\eta_{n+1} = T(\cdot,\nu_n)$ and $\eta^* = T(\cdot,\nu^*)$ for every saddle point, so Theorem A gives

$$m_n = \int_\Omega \varpi(x;\nu_n(x),\nu^*(x))\,dx \to 0. \tag{2.10}$$

Passing from weak convergence and (2.10) to strong convergence is the classical monotonicity argument for Leray–Lions operators. Pointwise, by the Cauchy–Schwarz inequality,

$$\varpi(x;a,b) \geq \big(g(x,|a|) - g(x,|b|)\big)\big(|a|-|b|\big) \geq 0. \tag{2.11}$$

If $g(x,\cdot)$ is strictly increasing and unbounded, (2.11) shows that $\varpi(x;a,b) \to \infty$ as $|a| \to \infty$ and that $\varpi(x;a,b) = 0$ only if $a = b$ (equality forces $|a| = |b|$ and $T(x,a)\cdot b = g(x,|a|)|b|$, i.e. $a = b$). Hence, along a subsequence for which the integrand in (2.10) tends to zero a.e., $\nu_n \to \nu^*$ a.e.; this is the argument of Leray and Lions [48], see also [15]. It requires neither growth conditions nor any regularity in x, but it only yields convergence in L^s for $s < 2$, through the L^2 bound and Vitali's theorem. Convergence in the natural energy space, convergence of the multipliers, and quantitative information need a quantitative version of the strict monotonicity. When $k(x,\cdot)$ is nondecreasing, which for $G = t^p/p$ means $p \geq 2$, the monotonicity gap controls $|a-b|^2$ from below:

Lemma 2.14. *If $t \mapsto k(x,t)$ is nondecreasing on $(0,\infty)$ and $k(x,0) := \lim_{t\downarrow 0} k(x,t)$, then $\varpi(x;a,b) \geq \frac{1}{2}k(x,\frac{|a-b|}{2})|a-b|^2$ for all a,b.*

Proof. With $k_a = k(x,|a|)$ and $k_b = k(x,|b|)$ we have $\varpi = \frac{1}{2}(k_a+k_b)|a-b|^2 + \frac{1}{2}(k_a-k_b)(|a|^2-|b|^2)$, and the second term is ≥ 0. Hence $\varpi \geq \frac{1}{2}k(x,\max\{|a|,|b|\})|a-b|^2$, and $\max\{|a|,|b|\} \geq |a-b|/2$. □

When $k(x,\cdot)$ decreases, as for $1 < p < 2$, the lower bound of ϖ degenerates where $|a|+|b|$ is large, and no lower bound $\varpi(x;a,b) \geq \theta(|a-b|)$ with $\theta(t) > 0$ for $t > 0$ can hold (cf. Proposition 2.9). For the p-Laplacian one compensates this with Hölder's inequality and the boundedness of the energy, and for the $p(x)$-Laplacian one usually splits Ω into the regions $\{p \geq 2\}$ and $\{p < 2\}$. For a general Young integrand there is no exponent with which to split Ω. The following lemma splits instead, at every point, according to the relative size of $|a-b|$ and $|a|+|b|$: where the difference is comparable to the vectors, the monotonicity gap controls it; where it is small compared with them, the energy does. It only uses Lieberman's condition, uniformly in x.

Lemma 2.15. *Let $G(x,\cdot) \in C^1([0,\infty)) \cap C^2(0,\infty)$, $g(x,0) = 0$, $g(x,t) > 0$ for $t > 0$, and assume that for some $0 < \delta \leq g_0 < \infty$*

$$\delta \leq \frac{t\,\partial_t g(x,t)}{g(x,t)} \leq g_0 \qquad \text{for a.e. } x \in \Omega \text{ and all } t > 0. \tag{L}$$

Put $\gamma = (g_0-1)_+$, $\beta = (1-\delta)_+$ and $c_1 = \min\{1,\delta\}\,4^{-1-\gamma}$. Then for a.e. x and all $a,b \in \mathbb{R}^d$, with $S = |a|+|b|$:

(i) $\varpi(x;a,b) \geq c_1\,k(x,S)\,|a-b|^2$;

(ii) *for every* $\lambda \in (0,1]$,

$$G(x,|a-b|) \le \frac{\lambda^{-\beta}}{c_1}\,\varpi(x;a,b) + 2^{1+g_0}\lambda^{1+\delta}\big(G(x,|a|)+G(x,|b|)\big);$$

in particular $G(x,|a-b|) \le c_1^{-1}\varpi(x;a,b)$ *if* $\delta \ge 1$*;*

(iii) *if* $g_0 < 1$, $|T(x,a)-T(x,b)| \le \frac{3}{\delta}\, g(x,|a-b|)$.

Proof. We fix x and omit it from the notation. From (L), $(\log g)' \in [\delta/t, g_0/t]$ and $(\log k)' \in [(\delta-1)/t, (g_0-1)/t]$; integrating, for $0 < t \le S$,

(2.12)
$$k(S)\Big(\frac{t}{S}\Big)^{\gamma} \le k(t) \le k(S)\Big(\frac{S}{t}\Big)^{\beta}, \qquad G(2t) \le 2^{1+g_0}G(t), \qquad (1+\delta)G(t) \le tg(t) \le (1+g_0)G(t),$$

the last two because $(1+\delta)g \le (tg)' \le (1+g_0)g$.

(i) For $\xi \ne 0$, $DT(\xi)$ is symmetric with eigenvalues $k(|\xi|)$ and $g'(|\xi|) \in [\delta k(|\xi|), g_0 k(|\xi|)]$. With $e = a-b$, $c_s = b+se$, $s \mapsto T(c_s)$ is absolutely continuous (its derivative is bounded by $\max\{1,g_0\}k(|c_s|)|e|$, integrable since $k(t) \le k(1)t^{\delta-1}$ for $t \le 1$), so $\varpi(a,b) = \int_0^1 e \cdot DT(c_s)e\,ds \ge \min\{1,\delta\}|e|^2 \int_0^1 k(|c_s|)\,ds$. If $|a| \ge |b|$, then $|c_s| \ge S/4$ for $s \in [\frac34, 1]$, and (2.12) gives $\int_0^1 k(|c_s|)\,ds \ge \frac14 k(S)4^{-\gamma}$; the other case is symmetric.

(ii) Let $e = a-b$, so that $|e| \le S$. If $|e| \ge \lambda S$, then $G(|e|) \le |e|\,g(|e|) = k(|e|)|e|^2 \le k(S)(S/|e|)^{\beta}|e|^2 \le \lambda^{-\beta}k(S)|e|^2$ by (2.12), and (i) gives $G(|e|) \le \lambda^{-\beta}\varpi/c_1$. If $|e| < \lambda S$, then, since $\log G$ grows at least like $(1+\delta)\log t$ by (2.12), $G(|e|) \le G(\lambda S) \le \lambda^{1+\delta}G(S)$, and $G(S) \le G(2\max\{|a|,|b|\}) \le 2^{1+g_0}(G(|a|)+G(|b|))$. In both cases the asserted inequality holds. If $\delta \ge 1$, then $\beta = 0$ and we let $\lambda \to 0$.

(iii) If $g_0 < 1$, then k is nonincreasing and $|DT| \le k$. Let s^* minimize $|c_s|$ over $\mathbb{R}$; then $|c_s| \ge |e||s-s^*|$ and $k(|c_s|) \le k(|e|)\max\{1, |s-s^*|^{\delta-1}\}$, so $|T(a)-T(b)| \le |e|\int_0^1 k(|c_s|)\,ds \le (1+\frac{2}{\delta})g(|e|) \le \frac{3}{\delta}g(|e|)$. □

Theorem B. *Assume* (S) *and that* $G(x,\cdot) \in C^1([0,\infty))$ *with* $g(x,0) = 0$ *for a.e.* x*; let* $(\bar u, \bar\nu, \bar\eta)$ *be the limit in Theorem A.*

(a) *If* $g(x,\cdot)$ *is strictly increasing with* $g(x,t) \to \infty$ *as* $t \to \infty$ *for a.e.* x*, then* $\nu_n \to \bar\nu$ *and* $\nabla u_n \to \nabla \bar u$ *strongly in* $L^s(\Omega;\mathbb{R}^d)$*, and* $u_n \to \bar u$ *in* $W_0^{1,s}(\Omega)$*, for every* $s \in [1,2)$.

(b) *If* $k(x,\cdot)$ *is nondecreasing for a.e.* x *and* $k_*(t) := \operatorname{ess\,inf}_x k(x,t) > 0$ *for every* $t > 0$*, then* $\nu_n \to \bar\nu$ *and* $\nabla u_n \to \nabla \bar u$ *strongly in* $L^2(\Omega;\mathbb{R}^d)$*, and* $u_n \to \bar u$ *in* $H_0^1(\Omega)$.

(c) *If* G *satisfies* (L)*, then, in addition to* (a), $\int_\Omega G(x,|\nu_n - \bar\nu|)\,dx \to 0$*, that is,* $\nu_n \to \bar\nu$ *in the Musielak–Orlicz space* $L^{G(\cdot)}$*. If moreover* $g_0 < 1$ *and* $K := \operatorname{ess\,sup}_x g(x,1) < \infty$*, then* $\int_\Omega G(x,|\nabla u_n - \nabla\bar u|)\,dx \to 0$ *and* $\eta_n \to \bar\eta$ *strongly in* $L^2(\Omega;\mathbb{R}^d)$.

Proof. Fix a saddle point (u^*,ν^*,η^*) and put $e_n = \nu_n - \nu^*$. In each part we show that $e_n \to 0$ in the stated sense; since $\nu_n \rightharpoonup \bar\nu$, this forces $\nu^* = \bar\nu$, and the assertions on $\nabla u_n = s_n + \nu_n$ follow from $s_n \to 0$ in L^2 (Theorem A) and Poincaré's inequality.

(a) Let $\varpi_n(x) = \varpi(x;\nu_n(x),\nu^*(x))$, so that $\varpi_n \to 0$ in L^1 by (2.10). Every subsequence has a further subsequence with $\varpi_n(x) \to 0$ for a.e. x. For such x, put $b = \nu^*(x)$; by (2.11) and $g(x,t) \to \infty$, the sequence $\nu_n(x)$ is bounded, and every limit point a satisfies $\varpi(x;a,b) = 0$ by continuity of $T(x,\cdot)$, hence $a = b$. Thus $e_n \to 0$ a.e. along the sub-subsequence. Since (e_n) is bounded in L^2, $(|e_n|^s)$ is uniformly integrable for $s < 2$, and Vitali's theorem gives $e_n \to 0$ in L^s along it; hence for the whole sequence.

(b) On $\{|e_n| > \tau\}$, Lemma 2.14 and the monotonicity of k give $\varpi(x;\nu_n,\nu^*) \ge \frac12 k_*(\tau/2)|e_n|^2$. Hence $\|e_n\|^2 \le \tau^2|\Omega| + 2m_n/k_*(\tau/2)$ for every $\tau > 0$, and $e_n \to 0$ in L^2 by (2.10).

(c) Condition (L) gives $\partial_t g > 0$ and $g(x,t) \ge g(x,1)t^{\delta}$ for $t \ge 1$, so (a) applies. By (2.6) and $F \ge 0 = F(0)$, $F(\nu_n) \le (\eta_{n+1},\nu_n) \le M := \sup_k\|\eta_{k+1}\|\sup_k\|\nu_k\| < \infty$, and $F(\nu^*) < \infty$. Integrating

Lemma 2.15(ii) with $a = \nu_n$ and $b = \nu^*$,

$$\int_\Omega G(x, |e_n|)\, dx \le \frac{\lambda^{-\beta}}{c_1}\, m_n + 2^{1+g_0}\lambda^{1+\delta}\big(M + F(\nu^*)\big) \qquad (0 < \lambda \le 1);$$

letting first $n \to \infty$ and then $\lambda \to 0$ gives $\int G(x, |e_n|) \to 0$, which is convergence in the Luxemburg norm because (L) implies the Δ_2 condition uniformly in x (2.12). If $g_0 < 1$, then $g(x,t) \le K(t^\delta + t^{g_0})$ by (L), and $G(x,t) \le tg(x,t) \le 2K(t^{1+\delta} + t^2)$. Since $\nu^* = \bar\nu$, $\eta^* = T(\cdot, \bar\nu)$ for every saddle point, so $\bar\eta = T(\cdot, \bar\nu)$; Lemma 2.15(iii) gives $|\eta_{n+1} - \bar\eta|^2 \le \frac{18K^2}{\delta^2}(|e_n|^{2\delta} + |e_n|^{2g_0})$, which is uniformly integrable and tends to 0 a.e. along sub-subsequences by (a), so $\eta_{n+1} \to \bar\eta$ in L^2, and $\eta_{n+1} - \eta_n = rs_n \to 0$. Finally $G(x, |s_n + e_n|) \le 2^{1+g_0}(G(x, |s_n|) + G(x, |e_n|))$, and $\int G(x, |s_n|) \le 2K \int (|s_n|^{1+\delta} + |s_n|^2) \to 0$. □

Remark 2.16 (The p-Laplacian with $1 < p < 2$: a rate in terms of m_n). For $G = t^p/p$, $1 < p < 2$, Lemma 2.15(i) is the classical inequality $\varpi(a,b) \ge c_p|a-b|^2(|a| + |b|)^{p-2}$, and Hölder's inequality with exponents $\frac{2}{p}$ and $\frac{2}{2-p}$ replaces the case distinction of Lemma 2.15(ii): with $S_n = |\nu_n| + |\nu^*|$,

$$\int_\Omega |e_n|^p = \int_\Omega \big(|e_n|^2 S_n^{p-2}\big)^{\frac{p}{2}} S_n^{\frac{p(2-p)}{2}} \le \Big(\frac{m_n}{c_p}\Big)^{\frac{p}{2}} \Big(\int_\Omega S_n^p\Big)^{\frac{2-p}{2}},$$

and $\int S_n^p$ is bounded by the energy bound in the proof of Theorem B(c). Hence $\|\nu_n - \bar\nu\|_{L^p}^2 \le C\, m_n$, and, since $\sum_n m_n < \infty$ by Theorem A, $\sum_n \|\nu_n - \bar\nu\|_{L^p}^2 < \infty$. For the $p(x)$-Laplacian the same computation, done on $\{p(x) < 2\}$ with Young's inequality instead of Hölder's, together with Lemma 2.14 on $\{p(x) \ge 2\}$, is the usual splitting argument; Lemma 2.15(ii) does the same thing without reference to an exponent.

Examples. Condition (L) holds, with *no continuity in* x, in three important cases: for $t^{p(x)}/p(x)$ with measurable p and $1 < p^- \le p \le p^+ < \infty$, with $\delta = p^- - 1$ and $g_0 = p^+ - 1$ (for constant $p \in (1,2)$, Theorem B(c) gives strong convergence in $W_0^{1,p}$ and convergence of η_n in L^2, and Remark 2.16 a quantitative form); for double phase and two-phase integrands $t^p/p + a(x)t^q/q$, $1 < p \le q < \infty$, with *any* measurable $a \ge 0$, with $\delta = p - 1$ and $g_0 = q - 1$; and for integrands of the type $t^p \log^\alpha(e + t)$. Thus, *whenever (S) holds*, the iteration converges strongly for the $p(x)$-Laplacian with $p^- < 2$, including Zhikov's checkerboard exponent, and for two-phase problems with $p < 2$ and discontinuous coefficients. None of these cases is covered by Glowinski's theorem; for $p \ge 2$, Lemma 2.8 applies. The exponential integrands fall outside (L) but are still covered. For $e^t - 1 - t$, which violates (L) because $tg'/g \to \infty$, Theorem B(b) applies, since $k = (e^t - 1)/t \ge 1$ is nondecreasing. For $e^t - 1$, where ∂F is multivalued, Glowinski's theorem applies via Lemma 2.8(a).

All of the above extends to nonhomogeneous boundary data by a standard shift.

Remark 2.17 (Nonhomogeneous Dirichlet data). If $u = \psi$ on $\partial\Omega$ with $\psi \in H^1(\Omega)$ and $F(\nabla\psi) < \infty$, write $u = \psi + v$, $v \in H_0^1$, and $\tilde F(q) = F(q + \nabla\psi)$. Then $(V, H, \nabla, \tilde F, \ell)$ satisfies (A1)–(A3), and Proposition 2.3 and Theorems A–B apply in the variables (v, q, η) with $\nu = q + \nabla\psi$; in the proof of Theorem B(c), $F(\nu_n) = \tilde F(q_n) \le \tilde F(0) + (\eta_{n+1}, q_n)$ is still bounded.

3. When does (S) hold, and what does it say about the Lavrentiev gap?

Theorems A–B reduce the convergence of the continuous iteration to Assumption (S), and Proposition 2.13 shows that nothing weaker suffices. Since (S) is a regularity property of the solution, the natural question is when it holds. This section first collects what is known about it, and then separates it from a different property, the absence of a Lavrentiev gap, which will govern the discrete problem in Section 5.

3.1. **Sufficient conditions, and the p-Laplacian.** The simplest way to verify (S) is to exhibit a sufficiently regular solution or an explicit flux.

Lemma 3.1 (Lipschitz solutions)**.** *Assume $G(x,\cdot) \in C^1([0,\infty))$, $g(x,0) = 0$ for a.e. x, and $\operatorname{ess\,sup}_x g(x,M) < \infty$ for every $M > 0$. If $u \in H^1_0(\Omega)$ has $\nabla u \in L^\infty$ and $\int_\Omega T(x,\nabla u)\cdot\nabla\varphi = \int_\Omega f\varphi$ for all $\varphi \in C^\infty_c(\Omega)$, then* (S) *holds with $(u, \nabla u, T(\cdot,\nabla u))$.*

Proof. The flux $\eta = T(\cdot,\nabla u)$ lies in $L^\infty \subset H$, the equation extends to V by density, $F(\nabla u) < \infty$, and $\eta \in \partial F(\nabla u)$ by Lemmas 2.4 and 2.2(iii). □

Lemma 3.2 (Explicit flux)**.** *Let $u \in H^1_0(\Omega)$ with $F(\nabla u) < \infty$ and $\eta \in H$ with $\eta(x) \in \partial\Phi(x,\cdot)(\nabla u(x))$ a.e. and $(\eta, \nabla w) = (f, w)$ for all $w \in H^1_0(\Omega)$ (e.g. $\eta \in W^{1,\infty}$ with $-\operatorname{div}\eta = f$). Then* (S) *holds.*

Proof. This follows from Lemma 2.4 and Proposition 2.3. □

Lemma 3.2 applies in particular to the radial problems used in our experiments, for which the flux is known in closed form.

Remark 3.3 (Radial test problems)*.* On the unit disk with $f = c > 0$ and a radial Young integrand $G(x,t) = G_0(|x|,t)$ such that $g_0(s,\cdot)$ is continuous and strictly increasing on $(0,\infty)$ and $\inf_s g_0(s,t) \to \infty$ as $t \to \infty$, let $\tau(s) = 0$ if $cs/2 \le g_0(s,0)$ and otherwise $g_0(s,\tau(s)) = cs/2$. Then $u(x) = \int_{|x|}^1 \tau(s)\,ds$ is Lipschitz, and $\eta(x) = -cx/2$ satisfies $-\operatorname{div}\eta = c$ and $\eta \in \partial\Phi(x,\cdot)(\nabla u)$ a.e., by Lemma 2.2(i) in the dead core of $e^t - 1$ and by Lemma 2.2(ii) elsewhere. By Lemma 3.2, (S) holds for these data. This includes $e^t - 1 - t$, $e^t - 1$, variable exponents with $p^- < 2$ and discontinuous radial coefficients, all of which lie outside the regularity theory reviewed below. The same holds for manufactured smooth solutions (Remark 2.17).

For the p-Laplacian, (S) can be characterized completely. The proof relies on a multiplier lemma, which is a form of the subdifferential sum rule.

Lemma 3.4 (Multiplier lemma)**.** *Let X be a real Hilbert space, $R \subset X$ a closed subspace, and $\Psi\colon X \to \mathbb{R}$ convex and bounded above on a ball centered at $q_0 \in R$. If q_0 minimizes Ψ over R, there is $\zeta \in \partial\Psi(q_0)$ with $\zeta \perp R$.*

Proof. By (P1), Ψ is continuous at q_0. The indicator ι_R of R has $\partial\iota_R(q_0) = R^\perp$, and q_0 minimizes $\Psi + \iota_R$, so (P4) gives $0 \in \partial\Psi(q_0) + R^\perp$. □

Proposition 3.5. *Let $G(t) = t^p/p$, $d \ge 2$ and $p \ge 2d/(d+2)$ (so that $W^{1,p}_0 \hookrightarrow L^2$), and let u_p be the minimizer of J over $W^{1,p}_0(\Omega)$.*

(i) *If $1 < p \le 2$:* (S) *holds $\iff$ J attains its minimum on $H^1_0(\Omega)$ $\iff$ $u_p \in H^1_0(\Omega)$.*
(ii) *If $p > 2$ and $\nabla u_p \in L^{2p-2}$, then* (S) *holds. Conversely, on a Lipschitz domain,* (S) *implies $\nabla u_p \in L^{2p-2}$.*

Proof. (i) If J attains its minimum on V at u, apply Lemma 3.4 with $X = H$, $R = \nabla(V)$, $q_\ell \in R$ representing $\nabla w \mapsto (f,w)$, $\Psi(q) = F(q) - (q_\ell, q)$ (finite and bounded on balls, as $|q|^p \le 1 + |q|^2$) and $q_0 = \nabla u$; $\eta = \zeta + q_\ell$ satisfies (2.1). Since J is continuous on $W^{1,p}_0$ and V is dense in it, a minimizer on V minimizes on $W^{1,p}_0$, so $u = u_p$; the converse implications are clear. (ii) If $\nabla u_p \in L^{2p-2}$, the flux is in L^2 and the Euler–Lagrange equation extends to V. Conversely, a saddle point has $\nabla u^* \in L^p$, so $u^* \in W^{1,p}_0$ (on a Lipschitz domain), $u^* = u_p$ and $|\nabla u_p|^{p-1} \in L^2$. □

The argument of (i) works verbatim whenever F is finite on L^2 (e.g. $G(x,t) \le A(x) + Ct^2$): then (S) holds if and only if J attains its minimum on H^1_0.

3.2. **What is known about (S).** For general integrands, (S) must be obtained from regularity theory. By Lemma 3.1, a gradient bound *up to the boundary* gives (S), whereas interior regularity, however strong, does not. We review the available results, from the best understood case, autonomous integrands, to the least understood one, energies of exponential growth.

Autonomous integrands with Lieberman's condition. This class includes t^p/p for every $1 < p < \infty$. Let $G = G(t)$ with $a(t) = g(t)/t \in C^1(0,\infty)$ and $-1 < i_a \le s_a < \infty$, where $i_a = \inf ta'/a$ and $s_a = \sup ta'/a$. This is condition (L) with $\delta = 1 + i_a$ and $g_0 = 1 + s_a$, the structure condition of Lieberman [50], which implies Δ_2 and ∇_2. Cianchi and Maz'ya [25, Thm. 2.3] prove that on every bounded convex $\Omega \subset \mathbb{R}^d$, $d \ge 2$, and for every $f \in L^2(\Omega)$, the generalized solution has flux $a(|\nabla u|)\nabla u \in W^{1,2}(\Omega)$, and [25, Thm. 2.4, Rem. 2.6] give the same conclusion on Lipschitz domains with $\partial\Omega \in W^{2,q}$, $q > 1$, if $d = 2$, or $\partial\Omega \in W^{2,d-1}$ if $d \ge 3$. Second-order regularity of the flux is enough for (S) in low dimensions, as the next result shows.

Proposition 3.6. *Under these hypotheses* (S) *holds if $d = 2$, or if $d \ge 3$ and $i_a \ge -2/d$ (for t^p/p: $p \ge 2 - 2/d$).*

Proof. The flux $\eta = a(|\nabla u|)\nabla u$ lies in $W^{1,2} \subset L^{2^*}$ (in every L^s, $s < \infty$, if $d = 2$). By (2.12), $g(t) \ge g(1)t^{1+i_a}$ for $t \ge 1$, so $|\nabla u| \le 1 + (|\eta|/g(1))^{1/(1+i_a)} \in L^2$ when $2^*(1 + i_a) \ge 2$. Then $u \in H^1_0$, the generalized solution is the weak one [25, Rem. 2.8], the equation extends to H^1_0 since $\eta \in L^2$, $F(\nabla u) \le \int |\nabla u||\eta| < \infty$, and $\eta \in \partial F(\nabla u)$. □

Together with [25, Thms. 2.3–2.4], Proposition 3.6 thus gives (S) for every $f \in L^2$ on bounded convex domains, if $d = 2$ or $i_a \ge -2/d$, and, when $d = 2$, also on Lipschitz domains with $\partial\Omega \in W^{2,q}$.

Several results give globally bounded gradients, and hence (S) by Lemma 3.1. Antonini [3, Cor. 1.4] (a preprint) treats $C^{1,\alpha}$ domains, $f \in L^\sigma$ with $\sigma > d$, $C^{1,\alpha}$ data and Hölder perturbations in x controlled by a *single* function b. Lieberman's boundary estimates [49, Thms. 1–2] were not available to us, but their role is confirmed by [50, p. 320], where their extension to condition (L) is also asserted. For $d \ge 3$, $f \in L^{d,1}$ and $\partial\Omega \in W^2L^{d-1,1}$, bounded gradients are obtained in [23] (as recalled in [25, Rem. 2.6]; we have not checked the original). For convex domains and p-type a with $2 \le p < d$, they follow from [24, Thm. 3.5], which also gives $|\nabla u|^{p-1} \in L^2$ already for $f \in L^{2d/(d+2)}$ [24, Thm. 4.1(ii)]. Convexity cannot simply be dropped: the examples in [24, §1] show that gradient bounds may fail on domains close to convex ones. Interior $C^{1,\alpha}$ regularity, by contrast, is classical [33, 62], and is proved for (L) in [50, Thm. 1.7]; as noted above, it does not give (S).

Variable exponents. For $p \in C^{0,\mu}(\overline{\Omega})$ with $p^- > 1$ and $\partial\Omega \in C^{1,\beta}$, global $C^{1,\alpha}$ regularity of bounded weak solutions is due to Fan, see [40]. We could not check its hypotheses in the original, so in this case (S) holds only conditionally. Interior results are given in [28, 1]. For log-Hölder or merely measurable exponents no gradient bound is known, and (S) must be assumed; when it holds, Theorem B(c) gives strong convergence without any continuity of p.

Double phase and two-phase energies. For $a \in C^{0,\alpha}$ only *interior* gradient regularity is known, and mostly for $f \equiv 0$: under $q/p < 1 + \alpha/d$ [26, Thm. 1.2] and $q/p \le 1 + \alpha/d$ [9, Thm. 1.1], and for bounded minimizers under $q \le p + \alpha$ [27, 9]. A right-hand side is allowed in [30, Thm. 1.3], for $a \in W^{1,\sigma}$ with $\sigma > d$, and general Musielak–Orlicz energies without right-hand side are treated in [47]. For double phase energies, therefore, (S) must be assumed. For discontinuous a nothing is available, except explicit solutions (Remark 3.3).

Exponential growth. For $e^t - 1 - t$ (where $s_a = \infty$) and $e^t - 1$ (where $g(0) = 1$) the situation is worse still. The results of [30, Thms. 1.4–1.5] require exponents > 1 in the exponential, those of [25, 3] require $s_a < \infty$, and [65, p. 4] only remarks that their global method should extend to non-Δ_2 growth. No result known to us gives (S) for general data, and only explicit radial solutions are available.

Failure of (S).. Finally, (S) fails whenever the minimizer is not in H^1_0 (Proposition 3.5(i)), even without any Lavrentiev phenomenon (Example 3.9 below).

In summary, (S) is known to hold globally for autonomous integrands with Lieberman's condition (under the domain and data assumptions above) and, conditionally on [40], for Hölder continuous exponents, while for log-Hölder and measurable exponents, double phase, two-phase and exponential energies it must be assumed or verified on explicit solutions. Whenever (S) holds, Theorem B applies to all of these energies except the exponential ones, for which Theorem B(a)–(b) or Glowinski's theorem applies.

3.3. **(S) and the Lavrentiev gap.** The literature on gradient regularity and the literature on the Lavrentiev phenomenon overlap, since both regularity and density of smooth functions fail for the same kind of non-autonomous integrands, and it is tempting to identify "(S) fails" with "there is a gap". This subsection shows that they are different properties. Throughout, Ω is a bounded Lipschitz domain, $\ell(v) = \int_\Omega fv$ with $f \in L^d(\Omega)$ (so that ℓ is continuous on $W^{1,1}_0$), the boundary datum is zero (except in Proposition 3.12(b), where Lipschitz data ψ are used and the infima are taken over $\psi + \mathcal{W}$, $\psi + H^1_0$, $\psi + C^\infty_c$), and

$$\mathcal{W} = \{v \in W^{1,1}_0(\Omega) : F(\nabla v) < \infty\}, \qquad m_{\mathcal{W}} = \inf_{\mathcal{W}} J, \qquad m_{H^1} = \inf_{H^1_0} J, \qquad m_{\mathcal{H}} = \inf_{C^\infty_c(\Omega)} J.$$

Then $m_{\mathcal{W}} \le m_{H^1} \le m_{\mathcal{H}}$, and the *gap* is $m_{\mathcal{H}} - m_{\mathcal{W}} \ge 0$. If $\int_\Omega G(x,t)\,dx < \infty$ for every t, then $m_{\mathcal{H}}$ is also the infimum over $W^{1,\infty}_0$ (truncate at levels $\pm\varepsilon$, mollify, and use dominated convergence). By Theorem A, under (S) the continuous iteration converges to a minimizer of J on H^1_0, so the value it targets is m_{H^1}. The first question is therefore where m_{H^1} lies.

Proposition 3.7 (What (S) says about the infima)**.** *Assume* (S) *with* $(u^*, \nabla u^*, \eta^*)$*, and let* $D_0(\sigma) = -\int_\Omega \Phi^*(x,\sigma)\,dx$*.*

(i) $J(u^*) = m_{H^1} = \max\{D_0(\sigma) : \sigma \in L^2(\Omega;\mathbb{R}^d),\ -\operatorname{div}\sigma = f \text{ in } \mathcal{D}'(\Omega)\}$*, attained at* η^**.*
(ii) *If* $G(x,t) \le A(x) + Ct^2$ *with* $A \in L^1$ *(subquadratic growth), then* $J(u^*) = m_{\mathcal{H}}$*.*
(iii) *If* $G(x,t) \ge ct^2 - A(x)$ *with* $c > 0$*,* $A \in L^1$ *(superquadratic growth), then* $\mathcal{W} \subset H^1_0$ *and* $J(u^*) = m_{\mathcal{W}}$*.*

Proof. (i) For admissible $\sigma \in L^2$ and $v \in H^1_0$ with $F(\nabla v) < \infty$, the Fenchel–Young inequality $\Phi(x,\nabla v) + \Phi^*(x,\sigma) \ge \sigma \cdot \nabla v$ and $\int \sigma \cdot \nabla v = \int fv$ give $J(v) \ge D_0(\sigma)$, with equality for (u^*, η^*) since $\eta^* \in \partial\Phi(x, \nabla u^*)$ a.e. (ii) Here F is finite, convex and l.s.c. on L^2, hence continuous, so J is continuous on H^1_0, in which C^∞_c is dense. (iii) If $v \in \mathcal{W}$, then $\nabla v \in L^2$, and the zero extension of v lies in $W^{1,1}(\mathbb{R}^d)$ with gradient in L^2 and compact support, hence in $H^1(\mathbb{R}^d)$, and it vanishes outside Ω, which on a Lipschitz domain characterizes $H^1_0(\Omega)$. □

Thus a dual certificate $\eta^* \in L^2$ certifies the minimum over H^1_0, not over $\mathcal{W}$. In the subquadratic regime (ii) the continuous method aims at $m_{\mathcal{H}}$, which may differ from $m_{\mathcal{W}}$; in the superquadratic regime (iii) it aims at $m_{\mathcal{W}}$, and a gap would make conforming discretizations inconsistent with the continuous method. As far as we know, both situations are compatible with (S). The first one does occur, as the following example shows.

Example 3.8 ((S) with a gap)**.** Let $\Omega = (-1,1)^d$, $1 < p^- < p^+ < 2$, and let $p(\cdot)$ and the linear functional $S^\circ(w) = \int b^\circ \cdot \nabla w$ be those of [7, Thm. 29], so that $\inf_{\mathcal{W}}(F + S^\circ) < \inf_{C^\infty_c}(F + S^\circ) = 0$. Since $b^\circ \in L^{p'(\cdot)} \subset L^2$ and $S^\circ = 0$ on C^∞_c [7, Prop. 22], S° vanishes on H^1_0. As an element of H^{-1} the load is therefore 0, so (S) holds with $(0,0,0)$ and $m_{H^1} = m_{\mathcal{H}} = 0 > m_{\mathcal{W}}$. The continuous iteration and P1 both return 0, and only a nonconforming method can see the gap. The example is degenerate, since the load is not of the form $\int fv$; we know no example of (S) with a gap and $f \in L^d$.

Conversely, (S) can fail although there is no gap at all.

Example 3.9 (No (S), no gap)**.** Let $p=\frac{3}{2}$, $\Omega=B_1\subset\mathbb{R}^2$, $f(x)=|x|^{-3/2}$ and $u(x)=4\log(1/|x|)$. Then $\eta=|\nabla u|^{-1/2}\nabla u=-2|x|^{-3/2}x$, $|\eta|=2|x|^{-1/2}\in L^3$, $-\operatorname{div}\eta=f$ in $\mathcal{D}'(B_1)$ (the flux through ∂B_ε is $-4\pi\varepsilon^{1/2}\to 0$), $u\in W_0^{1,3/2}(B_1)$ and $f\in L^{6/5}\subset(W_0^{1,3/2})^*\cap H^{-1}$. Hence u minimizes J on $W_0^{1,3/2}$, with $J(u)=(\frac{1}{p}-1)\int|\nabla u|^{3/2}=-\frac{32\pi}{3}$. Since $\nabla u\notin L^2$, (S) fails (Proposition 3.5(i)). Since J is continuous on $W_0^{1,3/2}$, on the other hand, $m_{\mathcal{W}}=m_{\mathcal{H}}=J(u)$: there is no gap, and P1 and CR approximate the same minimizer. Note that $\eta\in L^2$; what fails is $u\in H_0^1$.

Although (S) does not detect a gap, bounded fluxes do, in the following sense. The first part of the next proposition generalizes the elementary observation that a Lipschitz $\mathcal{W}$-minimizer rules out a gap.

Proposition 3.10 (Flux certificates)**.** *Let $u\in\mathcal{W}$ and $\eta\in L^\infty(\Omega;\mathbb{R}^d)$ with $\eta(x)\in\partial\Phi(x,\cdot)(\nabla u(x))$ a.e. and $\int\eta\cdot\nabla\varphi=\int f\varphi$ for all $\varphi\in C_c^\infty(\Omega)$. Then $J(u)=m_{\mathcal{W}}$. If moreover $u\in W^{1,\infty}$, there is no gap. Consequently, if (L) holds, $\operatorname{ess\,sup}_x g(x,1)<\infty$ and $m_{\mathcal{H}}>m_{\mathcal{W}}$, the flux $T(\cdot,\nabla u_{\mathcal{H}})$ of the minimizer $u_{\mathcal{H}}$ of J on the closure of C_c^∞ in $W^{1,G(\cdot)}$ is unbounded.*

Proof. We integrate $\Phi(x,\nabla v)\geq\Phi(x,\nabla u)+\eta\cdot\nabla(v-u)$ and use $\int\eta\cdot\nabla w=\int fw$ for $w\in W_0^{1,1}$, which follows by density since both sides are $W^{1,1}$-continuous. For the last claim, note that under (L) the closure of C_c^∞ is reflexive, so J has a minimizer $u_{\mathcal{H}}$ there with $J(u_{\mathcal{H}})=m_{\mathcal{H}}$, and its Euler–Lagrange equation holds for $\varphi\in C_c^\infty$; a bounded flux would make $u_{\mathcal{H}}$ a $\mathcal{W}$-minimizer. □

When is there no gap? For autonomous energies with zero data on convex (or star-shaped) domains there is never a gap, even without any Δ_2 condition.

Proposition 3.11 (No gap for autonomous energies)**.** *Let $G=G(t)$ be any finite convex nondecreasing function with $G(0)=0$, $f\in L^d$, zero Dirichlet data, and Ω bounded and star-shaped with respect to a ball (e.g. convex). For every $v\in\mathcal{W}$ there are $\varphi_k\in C_c^\infty(\Omega)$ with $\varphi_k\to v$ in $W^{1,1}$ and $F(\nabla\varphi_k)\to F(\nabla v)$; in particular $m_{\mathcal{W}}=m_{\mathcal{H}}$.*

Proof. Let the ball be $B_\rho(0)$; then $\operatorname{dist}(t\Omega,\partial\Omega)\geq(1-t)\rho$ for $t\in(0,1)$. Extend v by 0 and set $w_t(x)=t\,v(x/t)$. Then $\nabla w_t(x)=\nabla v(x/t)$, $\operatorname{supp}w_t\subset\overline{t\Omega}$ and $\int G(|\nabla w_t|)=t^dF(\nabla v)$. For $\varepsilon<(1-t)\rho$, $\varphi=\rho_\varepsilon*w_t\in C_c^\infty(\Omega)$ and, by Jensen's inequality, $\int G(|\nabla\varphi|)\leq\int\rho_\varepsilon*G(|\nabla w_t|)\leq F(\nabla v)$. As $t\uparrow 1$ and $\varepsilon\downarrow 0$, $\varphi\to v$ in $W^{1,1}$, and Fatou's lemma gives $F(\nabla\varphi)\to F(\nabla v)$; finally, ℓ is $W^{1,1}$-continuous. □

A localized version of this argument (partition of unity, inward translations, and a convex-combination trick replacing Δ_2) extends Proposition 3.11 to bounded Lipschitz domains; we do not use this extension and only mention it. For two-phase energies with a discontinuous coefficient, in contrast, the answer depends on the geometry of the phases.

Proposition 3.12 (Two-phase energies)**.** *Let $\Phi(x,\xi)=\frac{1}{p}|\xi|^p+\mathbf{1}_E(x)\frac{1}{q}|\xi|^q$, $1<p<q$, $f\in L^d$.*

(a) *If $E\Subset\Omega$, or $\Omega\setminus\overline{E}\Subset\Omega$, with E, $\Omega\setminus\overline{E}$ and Ω Lipschitz (e.g. $E=\{\frac{1}{2}<|x|<1\}$ in the unit disk), then $m_{\mathcal{W}}=m_{\mathcal{H}}$.*

(b) *If $d=2$, $\Omega=(-1,1)^2$, $E=\{x_1x_2>0\}$ and $p<2<q$, there are Lipschitz boundary data (and $f=0$) with $m_{\mathcal{W}}<m_{\mathcal{H}}$.*

Proof. (a) Let $v\in\mathcal{W}$, so that $v|_E\in W^{1,q}(E)$. We extend $v|_E$ with a Stein extension operator bounded simultaneously in $W^{1,p}$ and $W^{1,q}$ (if $\Omega\setminus\overline{E}\Subset\Omega$, extend from $\mathbb{R}^d\setminus\overline{(\Omega\setminus\overline{E})}$ the function equal to v on E and 0 outside Ω; if $E\Subset\Omega$, extend and cut off), obtaining $w\in W_0^{1,q}(\Omega)$ with $w=v$ on E. Then $z=v-w\in W_0^{1,p}(\Omega\setminus\overline{E})$. Approximating w in $W^{1,q}$ by $C_c^\infty(\Omega)$ and z in $W^{1,p}$ by $C_c^\infty(\Omega\setminus\overline{E})$, the sum converges in energy. (b) Zhikov's function u (0-homogeneous, ∇u supported in $\{x_1x_2<0\}$, $|\nabla u|\sim|x|^{-1}$) and separating field b (supported in E, $|b|\sim|x|^{-1}$, $b\cdot\nabla u=0$, $\operatorname{div}b=0$ in $\mathcal{D}'$), as recalled in [7, §1.1], satisfy $u\in W^{1,\Phi}$ (since $p<2$) and $b\in L^{\Phi^*}$ (since $\Phi^*(x,s)\leq s^{q'}/q'$ on

E and $q' < 2$). For large t and $s = t^\theta$, $p-1 < \theta < q-1$, $F(tu) + F^*(sb) \le C(t^p + t^{\theta q'}) < ts$, which is condition (10) of [7]; [7, Thm. 28], applied to Zhikov's pair (whose properties we take from [7, §1.1] and [63]; not checked in the original), then gives boundary data $g = t\,u^\partial$ (the boundary values of tu) for which the $\mathcal{W}$- and $\mathcal{H}$-minimizers differ and their energies satisfy $F(h_\mathcal{W}) < F(h_\mathcal{H})$. □

The proof carries over to the configuration rotated by $\pi/4$, $E = \{|x_1| < |x_2|\}$, which is used in Section 6. A gap for the specific data λu_0 used there, however, is *not proved*; it is only inferred numerically ([8] and Section 6.5).

In the planar checkerboards of Proposition 3.12(b) and of Zhikov's $p(x)$ (with the exponent > 2 on E), there is moreover an obstruction to H^1. An $H^1(B_\rho)$ function whose restrictions to two closed sectors with vertex 0 have continuous representatives has the same limit at 0 from both sectors, since $\int |\nabla v|^2 \ge \int_0^\rho \frac{1}{r} \int_0^{2\pi} |\partial_\theta v|^2 \, d\theta \, dr$ and $\int_0 \frac{dr}{r} = \infty$. Functions of $H^1 \cap \mathcal{W}$ are continuous at 0 on the two components of $\overline{E}$ (by Morrey's theorem, since the exponent there is > 2), with equal limits, whereas the singular competitors of [63, 7] jump. If, as in Zhikov's analysis, the closure of smooth functions consists exactly of the functions of $\mathcal{W}$ without such a jump (we have not checked this in the original), then $m_{H^1} = m_\mathcal{H}$ there. In that case, under (S), the continuous method aims at the $\mathcal{H}$-minimizer, just like P1, and the $\mathcal{W}$-minimizer is not in H^1.

What is known about the gap. Combining Propositions 3.11 and 3.12 with the literature gives the following picture for the energies considered in this paper; "a gap may occur" means that $m_\mathcal{H} > m_\mathcal{W}$ for some data.

- *Autonomous energies*, including t^p/p, integrands with (L) and integrands without Δ_2 such as $e^t - 1 - t$ and $e^t - 1$: there is no gap for zero data on domains that are star-shaped with respect to a ball (Proposition 3.11).
- *Variable exponents*: there is no gap for log-Hölder exponents, in particular for $p \in C^{0,\mu}(\overline{\Omega})$, by [64] and [32, §4.6] (neither read by us; both are recalled in [7, p. 22]). A gap may occur for continuous exponents that are not log-Hölder, with modulus of continuity $(\log(e + 1/t))^{-\kappa}$, $\kappa < 1$ [7, Thm. 30]; for discontinuous exponents with any range $[p^-, p^+]$ [7, Thm. 29]; and for the planar checkerboard with $p_1 < 2 < p_2$ [63] (not read by us; recalled in [7, §1.1]).
- *Double phase energies* with $a \in C^{0,\alpha}$: there is no gap for $q \le p + \alpha \max\{1, \frac{p}{d}\}$, globally on Lipschitz domains and for boundary data in $W^{1,q}$ [20, Thm. 2.3]. A gap may occur for $q > p + \alpha \max\{1, \frac{p-1}{d-1}\}$ [7, Thm. 32], and in the earlier counterexamples of [39], with $p < d < d + \alpha < q$; see also [14].
- *Two-phase energies* with $a = \mathbf{1}_E$: there is no gap if E or $\Omega \setminus \overline{E}$ is compactly contained in Ω and the sets involved are Lipschitz (Proposition 3.12(a)), whereas a gap may occur for the planar checkerboard with $p < 2 < q$ (Proposition 3.12(b)).

Summary. The results of Sections 2 and 3 can be summarized as follows. The continuous iteration converges if and only if (S) holds, and this is the case if and only if the iterates are bounded (Theorem A and Proposition 2.13). Assumption (S) concerns H^1_0, and it can fail without any gap (Example 3.9). The gap, in contrast, concerns $\mathcal{H}$ versus $\mathcal{W}$, and in general (S) neither implies nor excludes it (Proposition 3.7 and Example 3.8). A gap does, however, force the flux of the $\mathcal{H}$-minimizer to be unbounded (Proposition 3.10); the discrete counterpart of this fact, a computable diagnostic, is Theorem 5.7. Whatever the status of (S), when there is a gap the relevant question is which discrete problem approximates $m_\mathcal{W}$. Answering it requires, first of all, a convergence theory for the discrete iteration that does not depend on regularity; this is the subject of the next section.

4. The discrete problems: convergence without regularity

We now turn to the iteration that is actually computed. This section introduces the P1 and CR discretizations, which share the flux space and the nonlinear step, and shows that on every mesh the discrete iteration converges for every r, without any regularity assumption (Theorem C).

4.1. **Two discretizations with the same flux space.** Let $\Omega \subset \mathbb{R}^2$ be a bounded polygon and $\mathcal{T}_h$ a conforming triangulation of Ω. (For the disk in Section 6, Ω is replaced by the inscribed polygon Ω_h; everything in this section concerns the triangulated polygon.) We denote by $\mathcal{E}_h^\circ$ and $\mathcal{E}_h^\partial$ the sets of interior and boundary edges, by m_E the midpoint of an edge E, and by x_T the barycenter of a triangle T. The flux space is

$$H_h = \{q\colon \Omega \to \mathbb{R}^2 :\ q|_T \text{ constant } \forall T \in \mathcal{T}_h\}, \qquad (q,q')_h = \sum_T |T|\, q_T \cdot q'_T,$$

and for u we use either the conforming space or the Crouzeix–Raviart space [29],

$$\begin{aligned} \mathrm{P1}_h &= \{v \in C(\overline{\Omega}) :\ v|_T \in \mathbb{P}_1\ \forall T\}, \qquad V_h^{\mathrm{P1}} = \{v \in \mathrm{P1}_h :\ v = 0 \text{ on } \partial\Omega\},\\ \mathrm{CR}_h &= \{v\colon v|_T \in \mathbb{P}_1\ \forall T,\ v \text{ continuous at } m_E\ \forall E \in \mathcal{E}_h^\circ\},\\ V_h^{\mathrm{CR}} &= \{v \in \mathrm{CR}_h :\ v(m_E) = 0\ \forall E \in \mathcal{E}_h^\partial\}. \end{aligned}$$

For $v \in \mathrm{CR}_h$, $\nabla_h v \in H_h$ denotes the elementwise gradient (which is the gradient on $\mathrm{P1}_h$), and $B_h = \nabla_h\colon V_h \to H_h$. We assume that $G(x,\cdot)$ is a Young function for *every* x, write $G_T = G(x_T,\cdot)$ and $\Phi_T(\xi) = G_T(|\xi|)$, and set

$$F_h(q) = \sum_T |T|\, G_T(|q_T|), \qquad J_h(v) = F_h(\nabla_h v) - \ell_h(v),$$

where ℓ_h is a linear functional on V_h. In the code, ℓ_h is given by the barycenter rule for P1 and by the edge-midpoint rule for CR; both rules give $\int_\Omega fv$ when f is constant and, for piecewise constant f, when f is evaluated inside each triangle. When the mesh *resolves* G, that is, when

$$G(x,\cdot) = G_T \quad \text{for a.e. } x \in T,\ \text{for every } T \in \mathcal{T}_h, \tag{4.1}$$

the discrete energy $F_h(\nabla_h v) = \int_\Omega G(x,|\nabla_h v|)\,dx$ is the exact energy of the broken gradient.

The algorithm (2.3)–(2.5) is applied to $(V_h, H_h, B_h, F_h, \ell_h)$. The flux space, the pointwise step (one scalar equation (2.8) per triangle, by Lemma 2.6) and the multiplier step are *identical* for P1 and CR; only the linear step changes. It reads $rA\,U_n = b_h + [(r\nu_{n-1} - \eta_n, \nabla_h\phi_i)_h]_i$, where ϕ_i runs over the nodal basis of V_h^{P1} (unknowns: interior vertices) or over the CR basis ϕ_E, $E \in \mathcal{E}_h^\circ$, $\phi_E(m_{E'}) = \delta_{EE'}$ (unknowns: interior edges), and A is the stiffness matrix, which is factorized once. The abstract theory of Section 2 applies to this setting as soon as ∇_h is injective.

Lemma 4.1 (∇_h is injective)**.** *$\|\nabla_h v\|_h$ is a norm on V_h^{P1} and on V_h^{CR}; hence A is symmetric positive definite and $(V_h, H_h, B_h, F_h, \ell_h)$ satisfies (A1)–(A3) with $\|v\|_{V_h} := \|\nabla_h v\|_h$, F_h finite, convex and continuous.*

Proof. The P1 case is classical. If $v \in V_h^{\mathrm{CR}}$ and $\nabla_h v = 0$, v is constant on each triangle; neighbors through an edge $E \in \mathcal{E}_h^\circ$ share the value at m_E, any two triangles are joined by such a chain (the polygon is connected), and the constant vanishes at boundary midpoints. Finally, F_h is a finite sum of finite convex functions of the blocks q_T. □

An h-uniform version of Lemma 4.1, namely a broken Poincaré–Friedrichs inequality, is not needed for Theorem C; it is needed only for the limit $h \to 0$, see [55, (4.4)], [18] and [8, Lemma 3].

For a nonhomogeneous boundary datum ψ, the discrete admissible sets are

$$\begin{aligned} \mathrm{P1}_{h,\psi} &= \{v \in \mathrm{P1}_h :\ v(z) = \psi(z) \text{ at the boundary vertices}\},\\ \mathrm{CR}_{h,\psi} &= \Big\{v \in \mathrm{CR}_h :\ v(m_E) = \bar{\psi}_E := \frac{1}{|E|}\int_E \psi\,ds\ \ \forall E \in \mathcal{E}_h^\partial\Big\}. \end{aligned}$$

With a lifting $\tilde{u}_h$ (prescribed boundary values, zero interior unknowns) one writes $u = \tilde{u}_h + v$, $v \in V_h$, and applies everything to $\tilde{F}_h(q) = F_h(q + \nabla_h\tilde{u}_h)$, which is finite, convex, nonnegative and superlinear when F_h is; this is enough, since (A2) only asks F to be proper and any initialization is admissible (Remark 2.12).

4.2. **Theorem C: unconditional convergence of the discrete iteration.** In finite dimensions, superlinearity of the integrand is enough to guarantee existence of discrete saddle points, and with them the discrete counterpart of (S).

Theorem C. *Let $V_h = V_h^{\mathrm{P1}}$ or $V_h = V_h^{\mathrm{CR}}$ (possibly with nonhomogeneous data as above), let $G(x,\cdot)$ be a Young function for every x, and assume*

$$\lim_{t\to\infty} G_T(t)/t = +\infty \qquad \text{for every } T \in \mathcal{T}_h. \tag{4.2}$$

Then:

(i) *J_h has a minimizer u_h in the discrete admissible set; it is unique if every G_T is strictly convex.*

(ii) *There is $\eta_h \in H_h$ such that $(u_h, \nabla_h u_h, \eta_h)$ is a saddle point of the discrete augmented Lagrangian for every $r > 0$; equivalently, $\eta_{h,T} \in \partial\Phi_T(\nabla_h u_h|_T)$ for every T and $(\eta_h, \nabla_h w)_h = \ell_h(w)$ for every $w \in V_h$.*

(iii) *For every $r > 0$ and every initialization $\nu_0, \eta_1 \in H_h$, the iterates converge: $u_n \to \bar{u}_h$, $\nu_n \to \nabla_h \bar{u}_h$, $\eta_n \to \bar{\eta}_h$, where $(\bar{u}_h, \nabla_h \bar{u}_h, \bar{\eta}_h)$ is a discrete saddle point and $\bar{u}_h$ minimizes J_h; $\nabla_h u_n - \nu_n \to 0$, and the identity (2.9) holds with $(\cdot,\cdot)_h$, so that E_n is nonincreasing.*

Proof. By Lemma 4.1 and the lifting it suffices to treat V_h with $\tilde{F}_h$, which we still call F_h.

(i) Let L be the norm of ℓ_h on $(V_h, \|\nabla_h \cdot\|_h)$ and $a_{\min} = \min_T |T|$, so that $\|q\|_h \le a_{\min}^{-1/2} \sum_T |T| |q_T|$. Given $M > 0$, (4.2) gives $G_T(t) \ge M(t - t_M)$ for all $t \ge 0$ and T, hence $J_h(v) \ge (M a_{\min}^{1/2} - L)\|\nabla_h v\|_h - C_M$, also with the lifting since G_T is nondecreasing. Thus J_h is continuous and coercive on V_h and attains its minimum. If every G_T is strictly convex, then every G_T is strictly increasing, so Φ_T is strictly convex (for $|\xi_1| = |\xi_2|$, $\xi_1 \ne \xi_2$, use $|(\xi_1 + \xi_2)/2| < |\xi_1|$); hence F_h is strictly convex and so is $J_h = F_h \circ \nabla_h - \ell_h$, ∇_h being injective.

(ii) Since F_h is finite, $0 \in \partial J_h(u_h)$ and the chain rule in (P4) give $\eta_h \in \partial F_h(\nabla_h u_h)$ with $(\eta_h, \nabla_h w)_h = \ell_h(w)$ for all $w \in V_h$; this is the finite-dimensional existence of saddle points of [44, Ch. VI, §4.2], see also [12, Rem. 3.1]. Since F_h is a sum of functions of the blocks q_T with weights $|T| > 0$, $\eta_h \in \partial F_h(q)$ iff $\eta_{h,T} \in \partial\Phi_T(q_T)$ for all T. By Proposition 2.3, this is a saddle point for every $r > 0$.

(iii) By (ii), (S) holds for the discrete data, and Theorem A, whose proof uses only (A1)–(A3), gives weak convergence to a saddle point for any initialization (Remark 2.12); in finite dimensions, weak convergence is strong convergence. By Proposition 2.3, $\bar{u}_h$ minimizes J_h. □

Only two points in this argument depend on the space. The first is the injectivity of ∇_h (Lemma 4.1), where continuity at the midpoints of interior edges and vanishing at the midpoints of boundary edges replace the continuity of P1 functions; the second is the representation of the boundary datum, by edge means instead of nodal values. No discrete inf–sup condition and no h-uniform estimate is involved. Theorem C(iii) also follows from [37, Thm. 8], whereas Glowinski's finite-dimensional result [44, Ch. VI, Thm. 5.2] requires a strictly convex C^1 part. Superlinearity cannot be dropped: for $G(t) = t$ and large ℓ_h, J_h is unbounded below. On the other hand, Theorem C says nothing about the number of iterations or its dependence on h and r (see Section 6), nor about the limit $h \to 0$, which is the subject of Section 5.

For the lower bounds of Section 5 we need a multiplier that is equilibrated at every iteration, and not only in the limit. The linear step provides one.

Remark 4.2 (An exactly equilibrated multiplier). By (2.3), the field

$$\tilde{\eta}_n := \eta_n + r(\nabla_h u_n - \nu_{n-1}) = \eta_{n+1} + r(\nu_n - \nu_{n-1}) \in H_h \tag{4.3}$$

satisfies $(\tilde{\eta}_n, \nabla_h w)_h = \ell_h(w)$ for all $w \in V_h$ *at every iteration* (up to the round-off of the linear solve), whereas η_n itself is equilibrated only in the limit. By Theorem C(iii), $\tilde{\eta}_n \to \bar{\eta}_h$.

5. Consistency and the Lavrentiev phenomenon: a guaranteed bracket

Theorem C shows that, on every mesh, the discrete iteration converges. The remaining question is whether the discrete minima converge to the right value as $h \to 0$, and here the Lavrentiev gap, not (S), is decisive. In this section we first recall what is known without a gap, then show that P1 energies are upper bounds for $m_{\mathcal{H}}$ and converge to it, and finally prove that the CR iteration provides guaranteed lower bounds for $m_{\mathcal{W}}$ (Theorem D); together, these give a computable bracket for the gap. Throughout the section, $\Omega \subset \mathbb{R}^2$ is a polygon, $\psi \in W^{1,1}(\Omega)$ is the boundary datum, $f \in L^\infty(\Omega)$,

$$\mathcal{W}_\psi = \{v \in W^{1,1}(\Omega) : v - \psi \in W_0^{1,1}(\Omega)\}, \qquad \mathcal{H}_\psi = \mathcal{W}_\psi \cap W^{1,\infty}(\Omega),$$

$J(v) = \int_\Omega G(x, |\nabla v|)\,dx - \int_\Omega fv\,dx$ for $v \in \mathcal{W}_\psi$ (possibly $+\infty$), $m_{\mathcal{W}} = \inf_{\mathcal{W}_\psi} J$ and $m_{\mathcal{H}} = \inf_{\mathcal{H}_\psi} J$. The space $\mathcal{H}_\psi$ is the space $H = W \cap W^{1,\infty}$ of [8]; for $\psi = 0$ and G bounded on bounded sets, it gives the same $m_{\mathcal{H}}$ as the smooth functions of Section 3.3 (Remark 5.2).

5.1. **Without a gap.** Consider first autonomous, continuous and superlinear integrands on polygons, with $f \in L^\infty$, $W^{1,\infty}$ boundary data and uniformly shape-regular meshes. If there is no gap (which is always the case for scalar autonomous convex problems with zero data on convex domains, by Proposition 3.11), a subsequence of P1 minimizers converges weakly in $W^{1,1}$ to a $\mathcal{W}$-minimizer and the energies converge [55, Thm. 3.1]. Under the same hypotheses, CR minimizers with edge-mean data converge in the same sense without any assumption on the gap [55, Thm. 4.1], and strongly for strictly convex integrands. For integrands that are continuous in x and homogeneous data, conforming energies converge to the infimum over Lipschitz functions [22, Thm. 2.1]. Rates are known only under regularity assumptions on the solution: for the p-Laplacian [10, 52, 36], for $\Delta_2 \cap \nabla_2$ N-functions in the quasi-norm $\|\mathcal{V}(\nabla u) - \mathcal{V}(\nabla u_h)\|$ when $\mathcal{V}(\nabla u) \in W^{1,2}$ [35, 34], for Hölder and log-Hölder $p(\cdot)$ [17, 31], and for CR with Hölder $p(\cdot)$ [6]. None of these results applies to discontinuous coefficients or to $e^t - 1 - t$. For double phase energies with $a \in C^{0,\alpha}$ in the no-gap range of [20, Thm. 2.3] and boundary data in $W^{1,q}$, Proposition 5.1 below with a majorant sampling of a (Remark 5.2(c)) gives convergence of the P1 minima to $m_{\mathcal{W}}$; with barycenter sampling we have no proof.

5.2. **The conforming energy is an upper bound.** On meshes that resolve the coefficients, every P1 function is an admissible Lipschitz competitor, so P1 energies are upper bounds for $m_{\mathcal{H}}$; and as $h \to 0$ they converge to $m_{\mathcal{H}}$, not to $m_{\mathcal{W}}$.

Proposition 5.1 (P1: upper bound and limit)**.** *Assume (4.1), that $\psi|_{\partial\Omega}$ is affine on every boundary edge of $\mathcal{T}_h$, and that $\ell_h(v) = \int_\Omega fv$ for $v \in \mathrm{P1}_h$ (e.g. f piecewise constant and the barycenter rule, or $f = 0$).*

(i) *For every $v_h \in \mathrm{P1}_{h,\psi}$, $v_h \in \mathcal{H}_\psi$ and $J_h(v_h) = J(v_h) \geq m_{\mathcal{H}} \geq m_{\mathcal{W}}$. In particular this holds for every iterate u_n of the P1 iteration.*

(ii) *Let $(\mathcal{T}_h)$ be a shape-regular family of triangulations satisfying these assumptions for every h (e.g. the criss-cross meshes of Section 6, which resolve $|x_1| = |x_2|$), with maximal diameter $h \to 0$; assume $\sup_x G(x, M) < \infty$ for every $M > 0$ and $\mathcal{H}_\psi \neq \emptyset$. Then $\min_{\mathrm{P1}_{h,\psi}} J_h \to m_{\mathcal{H}}$.*

Proof. (i) The function v_h is Lipschitz, and on each boundary edge v_h and ψ are affine with equal endpoint values, so $v_h - \psi \in W_0^{1,1}(\Omega)$. By (4.1), $F_h(\nabla v_h) = \int_\Omega G(x, |\nabla v_h|)$.

(ii) By (i), $\liminf_h \min J_h \geq m_{\mathcal{H}}$. Let $v \in \mathcal{H}_\psi$ with $J(v) < \infty$, and let $I_h v \in \mathrm{P1}_{h,\psi}$ be its nodal interpolant. By [22, Lemma 2.1], $\|\nabla I_h v\|_{L^\infty} \leq C\|v\|_{W^{1,\infty}}$ with C depending only on the shape regularity; moreover $I_h v \to v$ uniformly and $\nabla I_h v \to \nabla v$ a.e. Dominated convergence, with the majorant $\sup_x G(x, C\|v\|_{W^{1,\infty}})$, gives $J(I_h v) \to J(v)$, so $\limsup_h \min J_h \leq J(v)$. □

Remark 5.2. (a) Under the assumptions of Proposition 5.1(ii) with $\psi \in W^{1,\infty}$, $m_{\mathcal{H}}$ equals the infimum over $\psi + C_c^\infty(\Omega)$: for $w = v - \psi$ Lipschitz with zero trace, $w_\varepsilon = (w - \varepsilon)_+ - (w + \varepsilon)_-$ has

compact support and $|\nabla w_\varepsilon| \le |\nabla w|$; mollify and use dominated convergence. (b) The result [22, Thm. 2.1] assumes an integrand continuous on $\overline{\Omega}\times\mathbb{R}^m\times\mathbb{R}^{m\times n}$ and homogeneous data, so it does not literally cover discontinuous $p(\cdot)$ or $a(\cdot)$; this is why we state Proposition 5.1, whose proof is that of [22]. (c) If (4.1) fails, (i) still holds for a *majorant* sampling, $G_T \ge G(x,\cdot)$ on T (e.g. $x_T \in \operatorname{argmax}_T a$ for $t^p/p + a(x)t^q/q$): then $J_h(v_h) \ge J(v_h) \ge m_{\mathcal{H}}$. If moreover $\sup_{x\in T}(G_T(t) - G(x,t)) \le \omega(h)\Gamma(t)$ with $\omega(h)\to 0$ and Γ bounded on bounded sets (e.g. a uniformly continuous), the proof of (ii) gives again $\min J_h \to m_{\mathcal{H}}$.

Thus, in the presence of a gap, the conforming method converges, but to the wrong value $m_{\mathcal{H}} > m_{\mathcal{W}}$; this is the failure of conforming finite elements described by Carstensen and Ortner [22]. Since the P1 iteration converges for every r (Theorem C), the failure is entirely one of consistency, and not of the solver. Balci, Ortner and Storn [8] showed that Crouzeix–Raviart elements avoid it. Since their assumptions matter for what follows, we recall them precisely.

Remark 5.3 (Convergence of CR energies to $m_{\mathcal{W}}$). The result [8, Thm. 1] assumes a Carathéodory integrand $\phi(x,\xi)$, convex in ξ, with $-c_0 + c_1|\xi|^{p_-} \le \phi(x,\xi) \le c_2|\xi|^{p_+} + c_0$, $1 < p_- < p_+ < \infty$, a polygon, $\psi \in W^{1,p_+}(\Omega)$, $f \in L^{p'_-}$, CR boundary values equal to edge means, the one-point quadrature $\phi(x_T,\cdot)$ on T, and either their conditions (A1)–(A3) or (B); it concludes that the discrete CR energies converge to $\min_{\mathcal{W}_\psi} J$, with convergence of the minimizers (strong for strictly convex integrands). For the checkerboard $p(\cdot)$ with $(p_-,p_+) = (3/2,3)$ and for the two-phase integrand $t^{3/2}/(3/2) + \mathbf{1}_{\{|x_1|<|x_2|\}}t^3/3$ on meshes satisfying (4.1), we have $\phi_h = \phi$ a.e., so (B) and (A1) with $c = 1$ hold trivially, (A2) holds with $C = 8$, and the growth condition holds with $c_1 = 1/3$, $c_2 = c_0 = 1$; the datum λu_0 of Section 6.4 has the trace of $\psi = \lambda x_2 \in W^{1,\infty}$. These are Experiments 1 and 3 of [8], where the authors state that the assumptions hold. Integrands of exponential growth violate the upper bound, so [8, Thm. 1] does not apply to them; for *autonomous* convex superlinear integrands, including $e^t - 1 - t$, the convergence of CR energies is [55, Thm. 4.1] (with $f \in L^\infty$, $W^{1,\infty}$ boundary data and uniformly shape-regular meshes).

Combined with Theorem C, Remark 5.3 gives Glowinski's method with CR elements a convergence theory in the Lavrentiev regime, *under the assumptions of* [8, Thm. 1], that is, polynomial two-sided growth and resolved coefficients or minimum quadrature. On every mesh the iteration converges, for every r, to the CR minimizer, and the CR minima converge to $m_{\mathcal{W}}$ as $h \to 0$. This theory gives no rates and says nothing about joint limits in the number of iterations and h, and it excludes exponential integrands. Two practical questions remain: how can one tell from computations whether there is a gap, and how far is a computed energy from $m_{\mathcal{W}}$? A lower CR energy alone answers neither, since the CR energy is lower also without a gap (Remark 5.6(e)). The next theorem gives a quantitative answer.

5.3. **The Crouzeix–Raviart energy and a guaranteed lower bound.** The lower bound comes from convex duality, with Raviart–Thomas fields as dual variables. By (P3) and (P5), for $\phi(\xi) = \gamma(|\xi|)$ with γ a Young function, $\gamma(|\xi|) + \gamma^*(|\zeta|) \ge \zeta\cdot\xi$, with equality if and only if $\zeta \in \partial\phi(\xi)$; if γ is superlinear, γ^* is finite, convex and continuous. Let

$$\mathrm{RT}_0(\mathcal{T}_h) = \{\sigma \in H(\mathrm{div},\Omega) :\ \sigma|_T(x) = a_T + b_T\,x,\ a_T \in \mathbb{R}^2,\ b_T \in \mathbb{R}\ \ \forall T\}$$

be the lowest-order Raviart–Thomas space [57]; for $\sigma \in \mathrm{RT}_0(\mathcal{T}_h)$, the normal component $(\sigma\cdot n)_E$ is constant on each edge. The first step is to turn a discretely equilibrated, piecewise constant CR flux into an exactly equilibrated Raviart–Thomas field.

Lemma 5.4 (Generalized Marini identity)**.** *Let f_h be piecewise constant on $\mathcal{T}_h$ and let $\zeta \in H_h$ satisfy*

$$(\zeta, \nabla_h w)_h = \int_\Omega f_h w\,dx \qquad \forall w \in V_h^{\mathrm{CR}}. \tag{5.1}$$

Then $\sigma := \zeta - \frac{f_h}{2}(x - x_T)$ *on each* T *belongs to* $\mathrm{RT}_0(\mathcal{T}_h)$ *and* $\operatorname{div}\sigma = -f_h$. *If* $f_h = 0$, $\sigma = \zeta$.

Proof. This is the generalized Marini formula of [6, Prop. 2.37(ii)], stated there for the flux of the discrete solution of a $p(\cdot)$-problem; its proof uses only the discrete equation, here (5.1), and applies verbatim. Directly: Green's formula on $T \ni E$ with the CR basis function ϕ_E gives $|E|(\sigma|_T \cdot n_T)|_E = |T|\zeta_T \cdot \nabla\phi_E|_T - f_T|T|/3$, so the jump of $\sigma \cdot n$ across $E \in \mathcal{E}_h^\circ$ is the residual of (5.1) for $w = \phi_E$. □

For homogeneous Dirichlet data and the exact discrete flux of a $p(\cdot)$-problem, the identity of Theorem D(ii) below is the discrete strong duality of [6, Prop. 2.37(iii)], whose discrete dual uses $G_T^*(|\Pi_T\sigma|)$ (the two coincide when $f = 0$). For an autonomous integrand of quadratic growth, the equivalence of the CR method with the Raviart–Thomas mixed method with one-point quadrature and the absence of a discrete duality gap were proved by Carstensen and Liu [21, Thm. 3.1], who also bound the CR energy from above by the exact energy up to a data-oscillation term [21, (4.9)]. Theorem D extends the duality to x-dependent integrands without growth conditions and with nonhomogeneous boundary data, and turns it into a guaranteed lower bound for $m_{\mathcal{W}}$ that remains valid in the presence of a Lavrentiev gap. We refer to [11] for general convex problems and to [53, 5] for the linear case. What is new here, as far as we know, is the comparison with the *continuous* infimum $m_{\mathcal{W}}$, which holds also in the presence of a Lavrentiev gap, together with the boundary term for edge-mean data, the certificate at every iterate of the ADMM, and the bracket with the P1 energy.

Theorem D (Guaranteed lower bound)**.** *Let* Ω *be a polygon triangulated by* $\mathcal{T}_h$ *with* (4.1), $\psi \in W^{1,1}(\Omega)$, $f \in L^\infty(\Omega)$. *For* $\sigma \in \mathrm{RT}_0(\mathcal{T}_h)$ *with* $\operatorname{div}\sigma = -f$ *define*

$$D(\sigma) = \sum_{E \in \mathcal{E}_h^\partial} (\sigma \cdot n)_E \int_E \psi \, ds - \sum_{T \in \mathcal{T}_h} \int_T G_T^*(|\sigma(x)|)\, dx \ \in [-\infty, \infty) \tag{5.2}$$

(*such* σ *exist only if* f *is piecewise constant on* $\mathcal{T}_h$).

(i) (Weak duality.) $J(v) \geq D(\sigma)$ *for every* $v \in \mathcal{W}_\psi$; *hence* $m_{\mathcal{W}} \geq D(\sigma)$. *If moreover* f *is piecewise constant and* $\ell_h(v) = \int fv$ *on* CR_h, *then* $J_h(v_h) \geq D(\sigma)$ *for every* $v_h \in \mathrm{CR}_{h,\psi}$.

(ii) (The CR minimum.) *Let* $f = f_h$ *be piecewise constant,* $\ell_h(v) = \int_\Omega f_h v$, *every* G_T *superlinear,* u_h *a minimizer of* J_h *on* $\mathrm{CR}_{h,\psi}$, η_h *as in Theorem C(ii), and* σ_h *the field of Lemma 5.4 with* $\zeta = \eta_h$. *Then*

$$J_h(u_h) - D(\sigma_h) = \sum_T \Big[\int_T G_T^*(|\sigma_h|)\, dx - |T|\, G_T^*(|\eta_{h,T}|) \Big] \ \geq 0.$$

In particular, if $f = 0$ *then* $\sigma_h = \eta_h$ *and*

$$\min_{\mathrm{CR}_{h,\psi}} F_h(\nabla_h \cdot) = F_h(\nabla_h u_h) = D(\eta_h) \leq m_{\mathcal{W}}.$$

(iii) (A certificate at every iteration.) *Under the assumptions of (ii), let* (u_n, ν_n, η_n) *be the CR iterates and* σ_n *the field of Lemma 5.4 with* $\zeta = \tilde{\eta}_n$ *from* (4.3). *Then* $D(\sigma_n) \leq m_{\mathcal{W}}$ *and* $D(\sigma_n) \leq J_h(u_n)$ *for every* n, *and* $D(\sigma_n) \to D(\bar{\sigma}_h)$, *the value in (ii) for the limit saddle point; if* $f = 0$, $D(\sigma_n) \to \min_{\mathrm{CR}_{h,\psi}} F_h(\nabla_h \cdot)$.

No growth condition beyond (4.2) (*used only for existence in (ii)–(iii)*), *no* Δ_2 *condition and no absence of a gap are assumed.*

Proof. (i) Let $v \in \mathcal{W}_\psi$ with $J(v) < \infty$ and $\int G_T^*(|\sigma|) < \infty$. By (4.1) and the Fenchel–Young inequality, $G(x, |\nabla v|) \geq \sigma \cdot \nabla v - G_T^*(|\sigma|)$ a.e. in T, and all terms are integrable. Since $\sigma, \operatorname{div}\sigma \in L^\infty$, we have $\int_\Omega \sigma \cdot \nabla w = \int_\Omega fw$ for $w \in C_c^\infty(\Omega)$, hence for $w \in W_0^{1,1}(\Omega)$; taking $w = v - \psi$ gives

$$J(v) \geq \int_\Omega \big(\sigma \cdot \nabla\psi + \psi \operatorname{div}\sigma\big)\, dx - \int_\Omega G^*(x, |\sigma|)\, dx.$$

Green's formula on each triangle, the equality of the traces of ψ from both sides of interior edges and the continuity of $\sigma\cdot n$ give $\int_\Omega(\sigma\cdot\nabla\psi+\psi\,\mathrm{div}\,\sigma)=\sum_{E\in\mathcal{E}_h^\partial}(\sigma\cdot n)_E\int_E\psi$. For the discrete statement, let $v_h\in \mathrm{CR}_{h,\psi}$. By the Fenchel–Young inequality with $\Pi_T\sigma:=|T|^{-1}\int_T\sigma$ and by Jensen's inequality,

$$J_h(v_h)\geq\sum_T|T|\big(\Pi_T\sigma\cdot\nabla v_h|_T-G_T^*(|\Pi_T\sigma|)\big)-\int fv_h\geq\sum_T\int_T\sigma\cdot\nabla v_h-\int fv_h-\sum_T\int_T G_T^*(|\sigma|),$$

and the elementwise Green formula turns $\sum_T\int_T\sigma\cdot\nabla v_h-\int fv_h$ into $\sum_T\int_{\partial T}v_h\,\sigma\cdot n_T$. On an interior edge, $\sigma\cdot n$ is constant and continuous and $\int_E v_h|_{T_\pm}=|E|\,v_h(m_E)$ from both sides, so the contributions cancel; on a boundary edge, $\int_E v_h=|E|\,\bar\psi_E=\int_E\psi$.

(ii) By Theorem C(ii), η_h satisfies (5.1), so $\sigma_h\in\mathrm{RT}_0(\mathcal{T}_h)$, $\mathrm{div}\,\sigma_h=-f_h$ and $\Pi_T\sigma_h=\eta_{h,T}$. The computation in (i) with $v_h=u_h$, now with equality in Fenchel–Young ($\eta_{h,T}\in\partial\Phi_T(\nabla u_h|_T)$), gives $J_h(u_h)=\sum_{E\in\mathcal{E}_h^\partial}(\sigma_h\cdot n)_E\int_E\psi-\sum_T|T|G_T^*(|\eta_{h,T}|)$, which is the claim. The bracket is ≥ 0 by Jensen's inequality, and it vanishes if $f_h=0$.

(iii) By Remark 4.2, $\tilde\eta_n$ satisfies (5.1), so (i) applies to σ_n. By Theorem C(iii), $\tilde\eta_n\to\bar\eta_h$, hence $\sigma_n\to\bar\sigma_h$ uniformly, and $D(\sigma_n)\to D(\bar\sigma_h)$ by continuity of G_T^*. □

Combining Theorem D with the upper bound of Proposition 5.1 yields the bracket announced in the introduction.

Corollary 5.5 (A guaranteed bracket)**.** *Under the assumptions of Proposition 5.1(i) and Theorem D, for every $v_h\in\mathrm{P1}_{h,\psi}$ (e.g. any P1 iterate) and every $\sigma\in\mathrm{RT}_0(\mathcal{T}_h')$ with $\mathrm{div}\,\sigma=-f$ on any triangulation $\mathcal{T}_h'$ satisfying (4.1) (e.g. σ_n from any CR iterate),*

$$D(\sigma)\;\leq\;m_{\mathcal{W}}\;\leq\;m_{\mathcal{H}}\;\leq\;J(v_h),\qquad\textit{hence}\qquad 0\leq m_{\mathcal{H}}-m_{\mathcal{W}}\leq J(v_h)-D(\sigma).$$

If moreover $f=0$ and the hypotheses of [8, Thm. 1] *and of Proposition 5.1(ii) hold, then*

$$\Delta_h:=\min_{\mathrm{P1}_{h,\psi}}J_h-\min_{\mathrm{CR}_{h,\psi}}J_h\;\to\;m_{\mathcal{H}}-m_{\mathcal{W}}\qquad(h\to0).$$

Proof. The chain of inequalities follows from Proposition 5.1(i) and Theorem D(i). For the limit, $\min_{\mathrm{P1}_{h,\psi}}J_h\to m_{\mathcal{H}}$ by Proposition 5.1(ii) and $\min_{\mathrm{CR}_{h,\psi}}J_h\to m_{\mathcal{W}}$ by [8, Thm. 1]. □

Since the bracket is meant to be used as a gap indicator, it is important to separate what it proves from what it only suggests.

Remark 5.6 (A computable gap indicator: precise status). (a) *Proved:* the width $J(v_h)-D(\sigma)$ is, on every mesh and for every pair of iterates, a rigorous upper bound for the gap $m_{\mathcal{H}}-m_{\mathcal{W}}$ (up to floating-point arithmetic and the round-off of the linear solves, which only affects the equilibration of σ_n; its normal jumps can be monitored). If the width is below ε, the gap is below ε: *absence of a gap can be certified to any computed accuracy.* (b) *Proved, under the hypotheses of Corollary 5.5:* $\Delta_h\to m_{\mathcal{H}}-m_{\mathcal{W}}$, so $\Delta_h\to0$ if and only if there is no gap. (c) *Not available:* a positive lower bound for the gap. It would require an upper bound for $m_{\mathcal{W}}$ or a lower bound for $m_{\mathcal{H}}$, and the dual problem characterizes $m_{\mathcal{W}}$, not $m_{\mathcal{H}}$ [8, Lemma 6]. The presence of a gap is therefore only *inferred* from the behavior of Δ_h under refinement; no rate for Δ_h is proved, and extrapolations are heuristic. (d) *Variants.* If (4.1) fails but $\gamma_T\leq G(x,\cdot)$ on T (a *minorant* sampling, e.g. a replaced by $\min_T a$ for $t^p/p+a(x)t^q/q$), then $G^*(x,\cdot)\leq\gamma_T^*$ and Theorem D(i) holds with G_T^* replaced by γ_T^*; for $t^p/p+a(x)t^q/q$ with a replaced by $\min_T a$, the CR scheme with integrand γ_T is a minimum quadrature in the sense of [8, §4]; for variable exponents, sampling at $\mathrm{argmin}_T\,p$ is a minorant only for large t, and a valid choice is $\gamma_T^*(s)=\max_{p\in[\min_T p,\max_T p]}s^{p'}/p'$. If the CR boundary values are not the exact edge means, $D(\sigma_n)$ computed with the exact $\int_E\psi$ remains a valid lower bound. For $f\neq0$, $x\mapsto G_T^*(|\sigma(x)|)$ is convex on T, so replacing $\int_T G_T^*(|\sigma|)$ by the vertex rule $\frac{|T|}{3}\sum_i G_T^*(|\sigma(a_i)|)$ keeps the bound rigorous; but then the CR energy itself is *not* shown to be a lower bound. (e) The

CR energy lies *below* $m_{\mathcal{W}}$ also without a gap (Theorem D(ii)), so the fact that CR has the lower energy is never, by itself, evidence of a gap.

5.4. **A second diagnostic: the size of discrete saddle points.** The bracket measures energies. The proof of Theorem C suggests a second computable quantity, the size of the discrete saddle point, which is also the initial value of the Lyapunov functional. The next theorem shows that its behavior under refinement detects both a failure of (S) and a gap; it is the discrete counterpart of Proposition 3.10.

Theorem 5.7 (Discrete diagnostics for (S) and the gap)**.** *Let $\{P_j\}$ be a fixed polygonal partition of the polygon Ω, $G(x,t) = G_j(t)$ on P_j, $(\mathcal{T}_h)$ shape-regular triangulations resolving the partition with $h \to 0$, $f \in L^\infty$ and exact loads $\ell_h = \ell$, and Lipschitz boundary data ψ with piecewise affine trace. Let $(u_h, \nabla u_h, \eta_h)$ be P1 discrete saddle points and $(u_h^{\mathrm{CR}}, \nabla_h u_h^{\mathrm{CR}}, \eta_h^{\mathrm{CR}})$ CR discrete saddle points (Theorem C), and let m_{H^1} be as in Section 3.3, with the datum ψ.*

(a) *If $\liminf_h(\|\nabla u_h\| + \|\eta_h\|) < \infty$, then* (S) *holds and $m_{\mathcal{H}} = m_{H^1}$.*

(b) *If $c|\xi|^{p_-} - c_0 \le \Phi(x,\xi) \le C|\xi|^{p_+} + c_0$ with $1 < p_- < p_+ < \infty$ and $\liminf_h(\|\nabla_h u_h^{\mathrm{CR}}\| + \|\eta_h^{\mathrm{CR}}\|) < \infty$, then* (S) *holds and $m_{H^1} = m_{\mathcal{W}}$.*

(c) *Hence: if* (S) *fails, the P1 family is unbounded in $L^2 \times L^2$, and so is the CR family under the growth assumption of (b); if $m_{\mathcal{H}} > m_{\mathcal{W}}$ and the growth assumption of (b) holds, at least one is; if moreover $G(x,t) \ge ct^2 - A$, the P1 family is.*

(d) *If* (L) *holds with $\operatorname{ess\,inf}_x g(x,1) > 0$ and $m_{\mathcal{H}} > m_{\mathcal{W}}$, then $\|\eta_h\|_{L^\infty} \to \infty$ for the P1 family.*

Proof. We write the proof for $\psi = 0$. For Lipschitz ψ with piecewise affine trace, we write $u_h \in I_h\psi + V_h$; then $(\eta_h, \nabla u_h) = \ell(u_h - I_h\psi) + (\eta_h, \nabla I_h\psi)$ and $J(u_h) = (\eta_h, \nabla I_h\psi) - \ell(I_h\psi) - \int \Phi^*(x,\eta_h)$, with $\nabla I_h\psi \to \nabla\psi$ in L^2, and the argument below goes through verbatim (for CR use edge means of ψ).

(a) We extract subsequences with $u_h \rightharpoonup u$ in H_0^1 and $\eta_h \rightharpoonup \eta$ in L^2. For $w \in C_c^\infty$, $\nabla I_h w \to \nabla w$ uniformly, so $(\eta, \nabla w) = \lim(\eta_h, \nabla I_h w) = \lim \ell(I_h w) = \ell(w)$, and by density for $w \in H_0^1$. For $q \in L^2$ with $F(q) < \infty$, let $\Pi_h q$ be the elementwise mean. Since G is constant in x on each element, Jensen's inequality gives $F(\Pi_h q) \le F(q)$, and the discrete subgradient inequality together with $(\eta_h, \Pi_h q) = (\eta_h, q)$ and $(\eta_h, \nabla u_h) = \ell(u_h)$ gives $F(q) \ge F(\nabla u_h) + (\eta_h, q) - \ell(u_h)$. Passing to the limit, by the weak l.s.c. of F and $\ell(u_h) \to \ell(u) = (\eta, \nabla u)$, we obtain $F(q) \ge F(\nabla u) + (\eta, q - \nabla u)$; thus $\eta \in \partial F(\nabla u)$, and (S) holds. Next, by the elementwise Fenchel equality, $J(u_h) = -\int \Phi^*(x,\eta_h) = D_0(\eta_h)$, and $J(u_h) \ge m_{\mathcal{H}}$ (Proposition 5.1(i)). By weak l.s.c. of $\int \Phi^*$, $m_{\mathcal{H}} \le \limsup D_0(\eta_h) \le D_0(\eta) = J(u) = m_{H^1}$ (Proposition 3.7(i)).

(b) By [55, Thm. 4.2], a compactness result for CR functions that does not depend on the integrand, a subsequence $u_h^{\mathrm{CR}} \to u$ in L^1 with $\nabla_h u_h^{\mathrm{CR}} \rightharpoonup \nabla u$ and $u \in W_0^{1,1}$; as $\nabla u \in L^2$, $u \in H_0^1$. The argument of (a), with the CR interpolant ($\nabla_h I^{\mathrm{CR}} w = \Pi_h \nabla w \to \nabla w$ uniformly), gives (S). By weak l.s.c. and [8, Thm. 1] (assumption (B) holds since $\phi_h = \phi$), $J(u) \le \liminf J_h(u_h^{\mathrm{CR}}) = m_{\mathcal{W}}$; as $u \in \mathcal{W}$, $J(u) = m_{\mathcal{W}}$, so $m_{H^1} = m_{\mathcal{W}}$.

(c) This follows from (a), (b) and Proposition 3.7(iii).

(d) Suppose that $\|\eta_h\|_{L^\infty} \le C$ along a subsequence. Then, since $|\eta_h| = g(x, |\nabla u_h|) \ge g(x,1)\min\{|\nabla u_h|^\delta, |\nabla u_h|^{g_0}\}$ on each element, $\|\nabla u_h\|_{L^\infty} \le C'$. By (a), (S) holds at a weak-$*$ limit (u,η) with $u \in W^{1,\infty}$, $\eta \in L^\infty$, and Proposition 3.10 gives $J(u) = m_{\mathcal{W}}$ with u Lipschitz: no gap. □

With $\nu_0 = \eta_1 = 0$, the quantity $\|\eta_h\|^2/(2r) + r\|\nabla u_h\|^2/2$ is the initial value E_0 of the Lyapunov functional of Theorem C. Part (c) therefore says that E_0 blows up under refinement for both discretizations when (S) fails (for CR under the growth assumption of (b)), and for at least one of them (for P1 if $G(x,t) \ge ct^2 - A$) when there is a gap. Unlike the bracket, however, Theorem 5.7 gives no quantitative information, and boundedness can only be observed, not certified, on a finite sequence of meshes.

6. Numerical experiments

The experiments have two aims, which mirror the two halves of the theory. The first is to show that the method behaves well across the whole range of integrands covered by Theorems A–C, including those outside Glowinski's theorem, and to explore what the theory does not cover: convergence orders, the influence of r, and the effect of a failure of (S) (Sections 6.2 and 6.3). The second is to use the bracket of Corollary 5.5 and the diagnostics of Theorem 5.7 on problems with a Lavrentiev gap (Sections 6.4–6.7).

6.1. **Implementation and protocol.** All computations use the Python package `glaplace`, with P1 or CR elements, piecewise constant ν and η, and one sparse LU factorization per problem. The pointwise step (2.8) is solved by Newton's method in $\tau = \log t$, which converges monotonically from an upper bound because $\tau \mapsto g(e^\tau) + re^\tau$ is convex and increasing; a bisection fallback is available, and on $2 \cdot 10^5$ random cases the two agree to a relative accuracy of 10^{-13}. All runs start from $\nu_0 = \eta_1 = 0$ and stop when *both* the scaled primal residual $\|\nabla_h u_n - \nu_n\|_h/(1 + \|\nabla_h u_n\|_h)$ and the discrete duality gap $J_h(u_n) - D_h(\tilde\eta_n)$, computed with the equilibrated field (4.3), are below given tolerances. For CR with $f = 0$, $D_h(\tilde\eta_n)$ is exactly the certificate $D(\sigma_n)$ of Theorem D(iii); for P1 it is only the discrete dual value. All results were produced with a single version of the code (source hash `ec5ad1de5d43`), with which the scripts and raw data are distributed. Unless stated otherwise, errors are true finite element errors, computed by high-order quadrature against exact solutions, and orders are least-squares fits over the three finest meshes. *All orders, iteration counts and extrapolations below are observations; no rate is proved.*

6.2. **Smooth problems: orders, the parameter r, and the Lyapunov functional.** *Radial torsion.* On the unit disk with $f = c$ and zero data, the exact solutions of Remark 3.3 are available for any radial integrand. We use unstructured meshes with target sizes $h = 0.2, \dots, 0.0125$ (88 to 23 227 nodes), $c = 1$, $r = 1$ and tolerance 10^{-9}; the only exception is $e^t - 1$, for which we take $c = 8$ and $r = 30$, so that the solution has a dead core $\{|x| \le 1/4\}$. Table 1 and Figure 1 show the results. In all cases the observed orders are 2 in $L^2(u)$ and 1 in $L^2(\nabla u)$, with two exceptions: the gradient order is 1.25 for the discontinuous radial exponent, and the orders are 1.90 and 0.98 for the two-phase case with unresolved interface. This includes $e^t - 1 - t$, which violates Δ_2, $t^{1.5}/1.5$, which is not covered by Glowinski's theorem, and variable exponents with $p^- < 2$, continuous or discontinuous across the circle $|x| = 1/2$ (which is resolved by the mesh). The nodal maximum error has a reduced order (1.1–1.4) for $t^4/4$, $t^2 \log(1 + t)$ and $e^t - 1$, which we attribute, without proof, to the limited smoothness of u at the origin, and for the two-phase coefficient $\mathbf{1}_{|x|>1/2}$ when its interface is not resolved by the mesh.

Some runs did not reach the residual tolerance, although their duality gaps were small. For $e^t - 1$, the two finest meshes did not reach it in 20 000 iterations, while the duality gap was below 10^{-9}; the computed solution is flat in the dead core ($|\nabla u_h| \le 2.2 \times 10^{-8}$ for $|x| < 0.2$). For t^p/p with $p = 1.2$ and $p = 1.1$, none of the eight runs reached it in 50 000 iterations (final residuals up to 2×10^{-8} and 4×10^{-7}, respectively), with duality gaps below 8×10^{-9}. The iteration slows down as $p \to 1$, where ∂F becomes multivalued in the limit.

For the radial variable exponents ($p^- < 2$), the discrete diagnostics $\|\nabla_h u_h\|$, $\|\eta_h\|$ and $\max|\eta_h|$ converge under refinement. This is consistent with (S), although Theorem 5.7 gives only the converse implication. On the finest mesh, the modular $\int |\nu_n - \nabla u_h|^{p_T}$ decays to 10^{-12}–10^{-16} in 1000 iterations, in line with Theorem B(c), which, however, concerns the continuous iteration. CR elements give the same orders, with slightly smaller errors and fewer iterations.

Manufactured solution. For a nonradial test, we take $(0, 1)^2$ with $G(x, t) = t^2/2 + (1 + x_1)t^4/4$, $u = e^{x_1+x_2}$, Dirichlet data u, $r = 100$ and tolerance 10^{-10}, the pairwise orders are 2.02, 2.00, 2.00 (L^2) and 1.00 (gradient).

TABLE 1. Smooth problems with exact solutions (P1): observed orders of the relative L^2 errors of u and ∇u and of the nodal maximum error, and iterations on the coarsest/finest mesh. Top and middle: radial problems on the unit disk (h down to 0.0125, or 0.025 for $p \to 1$); bottom: manufactured solution on $(0,1)^2$ ($n = 10, \dots, 80$, pairwise orders). "Theory": result giving convergence of the continuous iteration, given that (S) holds for these data (Remark 3.3); G: Glowinski's theorem via Lemma 2.8; B: Theorem B. *: residual tolerance not reached.

G	theory	$L^2(u)$	$L^2(\nabla u)$	max	iterations
$t^{1.5}/1.5$	B	2.01	1.02	1.94	147/837
$t^2/2$	G, B	2.00	1.02	2.02	29/29
$t^4/4$	G, B	2.00	1.00	1.17	47/176
$t^2/2 + t^4/4$	G, B	2.00	1.01	2.02	30/26
$t^{1.5}/1.5 + t^3/3$	G, B	2.01	1.02	1.98	148/843
$t^2 \log(1+t)$	G, B	2.00	1.01	1.37	36/60
$t^3/\log(1+t)$	G, B	2.00	1.02	2.02	41/33
$e^t - 1 - t$	G, B(b)	2.00	1.01	2.02	29/25
$t^2/2 + \mathbf{1}_{\lvert x\rvert>1/2} t^4/4$ (unresolved)	G, B	1.90	0.98	0.79	30/27
$e^t - 1$, $c = 8$, $r = 30$	G	1.97	1.01	1.11	252/20000*
$t^{1.2}/1.2$	B	1.98	1.03	–	50000*
$t^{1.1}/1.1$	B	1.84	1.02	–	50000*
$t^{p(x)}/p(x)$, $p = 1.4 + 0.4\lvert x\rvert$	B	2.03	1.01	–	195/3733
$t^{p(x)}/p(x)$, $p = 1.5 + \lvert x\rvert$	B	2.03	1.02	–	59/569
$t^{p(x)}/p(x)$, $p = 1.5$ for $\lvert x\rvert < \frac{1}{2}$, 3 else	B	2.03	1.25	–	125/623
$t^2/2 + (1+x_1)t^4/4$, $u = e^{x_1+x_2}$	G, B	2.00	1.00	–	163/218

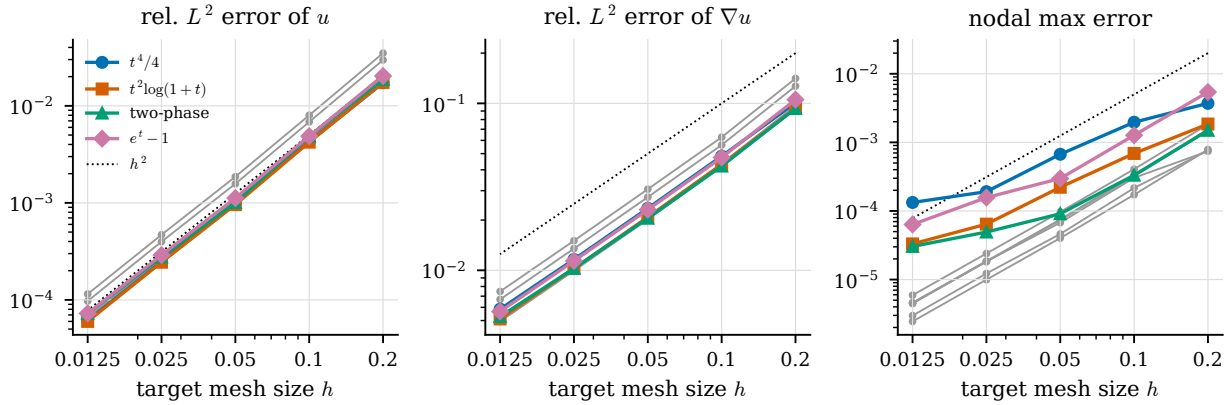


FIGURE 1. Radial torsion on the unit disk (P1, $r = 1$): relative L^2 errors of u (left) and ∇u (center) and nodal maximum error (right) versus the mesh size. Colored: the four cases with reduced maximum-norm order ($t^4/4$, $t^2\log(1+t)$, two-phase with unresolved interface, $e^t - 1$); gray: the other integrands in the top block of Table 1; dotted: reference slopes.

The parameter r. Theorem C guarantees convergence for every $r > 0$, but the speed of convergence depends strongly on r (Figure 2). On the checkerboard problems of Section 6.4, the iteration count varies by a factor of about 14 (P1) to 38 (CR) over $r \in [0.1, 100]$, with the best r near 1 for CR and between 1 and 3 for P1, and it grows moderately under refinement at fixed r. For the

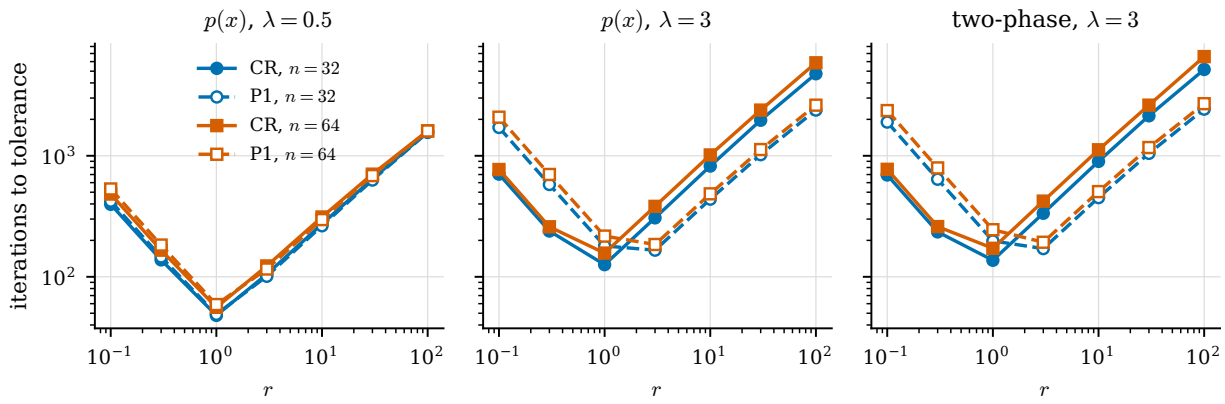


FIGURE 2. Iterations to tolerance versus the penalty parameter r, for CR (solid) and P1 (dashed) on criss-cross meshes with $n = 32$ and $n = 64$: $p(x)$ checkerboard with $\lambda = 0.5$ (no gap) and $\lambda = 3$, and two-phase checkerboard with $\lambda = 3$ (Section 6.4).

manufactured problem ($n = 40$), the counts for $r = 1, 10, 30, 100, 300$ are 11 575, 1167, 396, 208 and 516, with L^2 errors that agree to four digits. For the dead core problem ($e^t - 1$, $h = 0.05$, tolerance 10^{-8}), the counts for $r = 1, 3, 10, 30, 100$ are 6403, 2139, 650, 244 and 529; for $r = 100$, however, the error at termination is 1.1292×10^{-3}, against 1.1204×10^{-3} for $r \le 10$. For large r, then, a small primal residual does not imply a small iteration error. The duality gap (1.3×10^{-8} against 2.1×10^{-9}) and the dual residual do flag this run, which is why we stop on both the residual and the gap.

The Lyapunov functional. Using a reference saddle point computed to residual 10^{-14}, we checked that E_n in (2.9) was nonincreasing at every iteration in all runs examined: CR on the $p(x)$ checkerboard ($\lambda = 3$, $n = 32$, $r = 1, 3, 10$, with $E_{100}/E_1 = 3 \times 10^{-14}$, 6×10^{-7} and 8×10^{-4}, respectively) and P1 on the disk ($h = 0.1$) for five integrands, including the nonsmooth $e^t - 1$. After a transient, the decay is geometric, with rates that depend on G and r; no rate is proved.

6.3. **(S) fails, but there is no gap.** What happens when (S) fails? Example 3.9 can be computed: we take $p = 3/2$ on the unit disk with $f = |x|^{-\beta}$. For $\beta = 1.5$, the exact solution is $4\log(1/|x|)$, $\nabla u \notin L^2$ and (S) fails; for $\beta = 1.4$, $|\nabla u| \sim |x|^{-0.8} \in L^2$ and (S) holds (Proposition 3.5(i)); for $\beta = 1.6$, (S) fails again. The singular load must be integrated accurately. We use a Gauss–Jacobi rule on the triangles at the origin; the barycenter rule gives an energy error 2.5 times larger and a far too small $\|\nabla u_h\|$. Figure 3 shows the diagnostics of Theorem 5.7. For $\beta = 1.5$, $\|\nabla_h u_h\|^2$ grows by about $32\pi \log 2 \approx 69.7$ per halving of h (observed: +67, +75, +61, +69), consistently with $\int_{|x|>h} |\nabla u|^2 = 32\pi \log(1/h)$, while $\|\eta_h\|^2 \to 8\pi = \|\eta\|^2$ stays bounded and the iteration counts grow ($195 \to 589$). For $\beta = 1.4$, in contrast, the increments of $\|\nabla_h u_h\|^2$ decrease.

The P1 and CR energies approach the same limit $J(u) = -32\pi/3$, slowly ($J(u_h) - J(u) = 7.58 \to 1.97$). P1 approaches it from above, since $u_h \in W_0^{1,3/2}$ (up to the quadrature of the singular load). Here the CR energies are also above $J(u)$. This does not contradict Theorem D, which does not apply: $\Omega_h \neq \Omega$, f is not piecewise constant, and the statement on the CR energy itself needs $f = 0$. Thus a failure of (S) is visible in the iteration and in the size of the discrete saddle points, but it is not a consistency problem: P1 is adequate.

6.4. **The Lavrentiev experiments: setting.** We now turn to problems with a Lavrentiev gap. We take $\Omega = (-1, 1)^2$, $f = 0$, and the boundary datum λu_0, where $u_0 = 1$ on $\{|x_1| < x_2\}$, $u_0 = -1$ on $\{|x_1| < -x_2\}$ and $u_0 = x_2/|x_1|$ elsewhere; its trace is λx_2. The integrands are the Zhikov-type

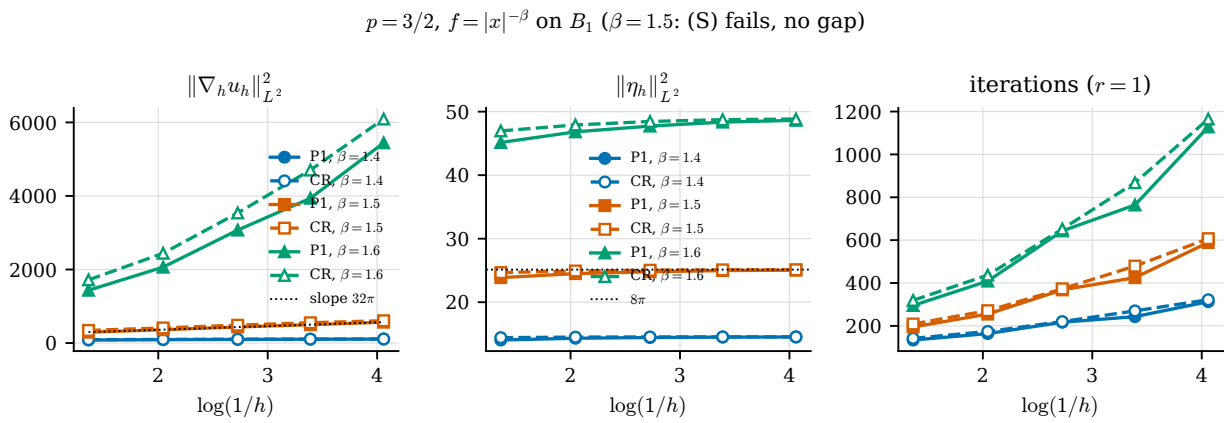


FIGURE 3. Failure of (S) without a gap: $G = t^{3/2}/(3/2)$, $f = |x|^{-\beta}$ on the unit disk, P1 (solid) and CR (dashed), $r = 1$. Squared L^2 norms of the discrete gradient (left) and flux (center), and iterations (right), versus $\log(1/h)$. (S) fails for $\beta = 1.5, 1.6$ and holds for $\beta = 1.4$; dotted: slope 32π and level 8π predicted for $\beta = 1.5$.

checkerboard exponent [63]

$$G(x,t) = t^{p(x)}/p(x), \qquad p = 3 \text{ on } \{|x_1| < |x_2|\}, \quad p = \tfrac{3}{2} \text{ elsewhere},$$

and the two-phase checkerboard $G(x,t) = t^{3/2}/(3/2) + \mathbf{1}_{\{|x_1|<|x_2|\}} t^3/3$. These are Experiments 1 and 3 of Balci, Ortner and Storn [8], who report, on the basis of numerical evidence, a gap for $\lambda > 1$ and for $\lambda \geq 0.4$, respectively. For $\lambda = 1$ and the $p(x)$ problem, the minimizer is x_2, with energy 2. We use criss-cross meshes ($n \times n$ squares, each cut into four triangles by its diagonals, $n = 8, \ldots, 512$), which resolve the lines $|x_1| = |x_2|$, so that (4.1) holds. P1 nodal data and CR edge means are exact, $r = 3$, and the tolerances are 10^{-10} for the residual and 10^{-9} for the relative duality gap. All 296 runs converged; the largest CR problem ($n = 512$, 1.57×10^6 unknowns) took 792 iterations.

By Proposition 5.1(i), Theorem D(ii) and Remark 5.3, all assumptions of Corollary 5.5 hold: every computed P1 energy is an upper bound for $m_{\mathcal{H}}$, every CR energy is a lower bound for $m_{\mathcal{W}}$, and $\Delta_h \to m_{\mathcal{H}} - m_{\mathcal{W}}$. The certificate is exact up to round-off. The normal jumps of σ_n never exceeded 1.1×10^{-13}, and $J_h(u_n) - D(\sigma_n)$ was never below -8×10^{-10} (relative 4×10^{-11}), so the bounds hold up to an absolute error of order 10^{-10}. The conjugates of the two-phase integrand, which have no closed form, are evaluated as rigorous upper bounds ($s\,t_{\mathrm{hi}} - G(t_{\mathrm{lo}})$ on a verified bracket), so their evaluation does not affect the direction of the bound. For $\lambda = 1$, both energies equal 2 to within 1.5×10^{-11} on every mesh.

6.5. **Brackets, widths and the signature of a gap.** Figure 4 shows the bracket for $\lambda = 3$, and Table 2 lists the finest-mesh values for a range of λ. Typical rigorous conclusions are

$$17.7635 \leq m_{\mathcal{W}} \leq m_{\mathcal{H}} \leq 18.9581 \quad (p(x),\ \lambda = 3),$$
$$20.9938 \leq m_{\mathcal{W}} \leq m_{\mathcal{H}} \leq 23.0881 \quad (\text{two-phase},\ \lambda = 3),$$

so that the gap is at most 1.195 and 2.094, respectively. At the other end of the range, the gap is at most 8.0×10^{-7} for $p(x)$ with $\lambda = 0.5$, and at most 2.5×10^{-8} for the two-phase problem with $\lambda = 0.2$. These numbers certify the (near) absence of a gap; the positivity of a gap, as explained in Remark 5.6(c), cannot be certified.

Figure 5 shows Δ_h versus the number of unknowns for all values of λ, and two regimes are clearly separated. For small λ the width closes at a steady rate. For large λ, in contrast, the closure ratio

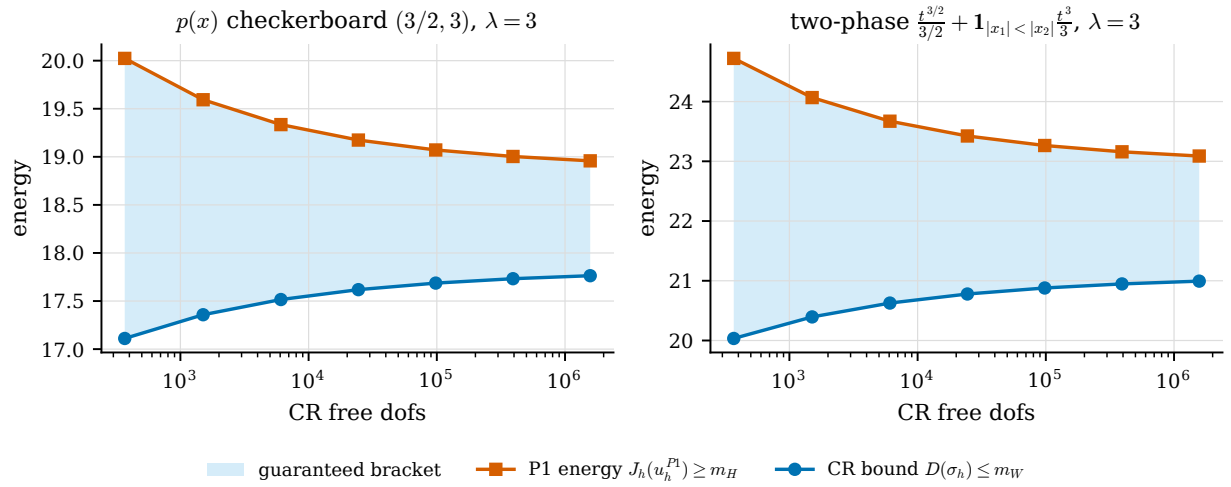


FIGURE 4. Guaranteed brackets $D(\sigma_h) \leq m_{\mathcal{W}} \leq m_{\mathcal{H}} \leq J_h(u_h^{\mathrm{P1}})$ (shaded) versus the number of CR unknowns, for the $p(x)$ (left) and two-phase (right) checkerboards with boundary datum $3u_0$. Upper curve: P1 energy; lower curve: CR energy $= D(\sigma_h)$.

TABLE 2. Finest-mesh brackets for the checkerboard problems: P1 energy ($\geq m_{\mathcal{H}}$), certified lower bound $D(\sigma_n)$ ($\leq m_{\mathcal{W}}$), width $\Delta_h \geq m_{\mathcal{H}} - m_{\mathcal{W}}$, ratio $\Delta_{n/2}/\Delta_n$ on the two finest meshes, and slope of Δ_h versus the number of unknowns (four finest meshes). Ratio 4, slope -1: $O(h^2)$ closure.

problem, λ	n	P1 energy	lower bound D	width Δ_h	ratio	slope
$p(x)$, 0.5	512	0.51381346	0.51381265	8.0×10^{-7}	4.00	−1.00
$p(x)$, 1	256	2.00000000	2.00000000	$\leq 1.5 \times 10^{-11}$	–	–
$p(x)$, 1.08	256	2.33326877	2.33325614	1.3×10^{-5}	2.02	−0.53
$p(x)$, 1.5	256	4.52197423	4.51556823	6.4×10^{-3}	1.31	−0.24
$p(x)$, 2	256	8.12513230	8.02915903	9.6×10^{-2}	1.17	−0.15
$p(x)$, 3	512	18.95812228	17.76348924	1.195	1.06	−0.06
$p(x)$, 4	256	35.76099252	30.51017011	5.251	1.07	−0.06
two-phase, 0.2	512	0.24363748	0.24363745	2.5×10^{-8}	3.23	−0.84
two-phase, 0.4	256	0.71309620	0.71309059	5.6×10^{-6}	2.45	−0.67
two-phase, 1	256	3.15984497	3.15462093	5.2×10^{-3}	1.34	−0.26
two-phase, 3	512	23.08812916	20.99380420	2.094	1.06	−0.06
two-phase, 4	256	41.75864138	34.43734359	7.321	1.06	−0.06

per refinement decreases towards 1 (1.30, 1.23, 1.17, 1.12, 1.09, 1.06 for $p(x)$ with $\lambda = 3$), that is, the width stalls. Since Δ_h converges to the gap by Corollary 5.5, a width that stalls at a value of order 1 is the signature of a gap, although it is not a proof of one. Heuristic extrapolations (Aitken's method, or least-squares fits with a free exponent over the three or four finest meshes) give closure exponents ≈ 0.56–0.60 in h that *decrease* with refinement. The free-exponent fits give estimated gaps of about 1.04 ($p(x)$, $\lambda = 3$) and 1.85 (two-phase, $\lambda = 3$), and fixed exponents 1 or 2 would give up to 1.17 and 2.06. Because the fitted exponent keeps drifting, we regard these numbers only as conjectural; the rigorous statement is the bracket.

No-gap controls, and a first surprise. For the $p(x)$ problem with $\lambda \in \{0.25, 0.5\}$, the ratios tend to 4.00 (slope -1.00): both bounds close at $O(h^2)$, as expected without a gap (compare the

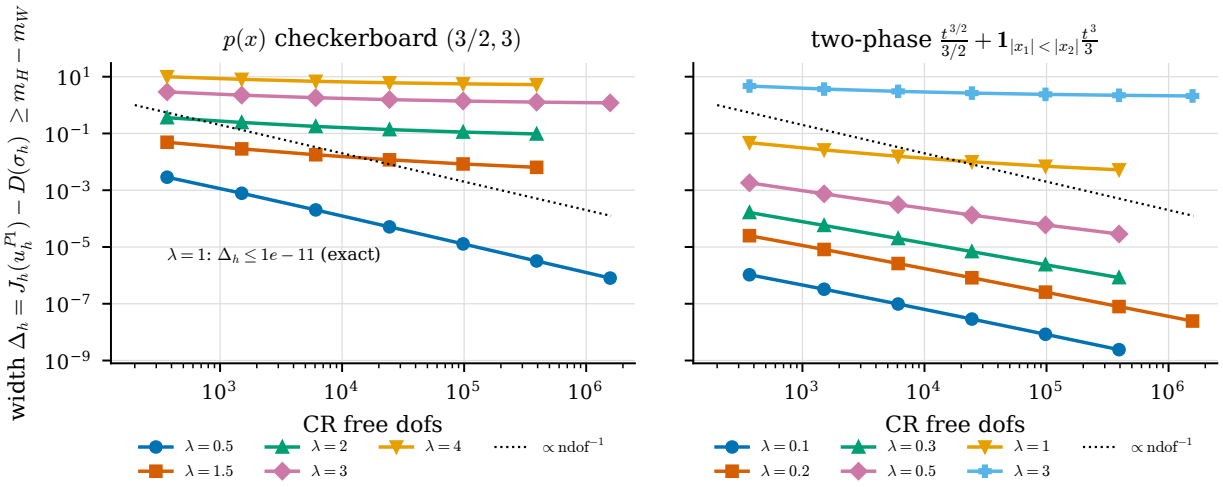


FIGURE 5. Width $\Delta_h = J_h(u_h^{\mathrm{P1}}) - D(\sigma_h) \geq m_{\mathcal{H}} - m_{\mathcal{W}}$ of the guaranteed bracket versus the number of CR unknowns, for the $p(x)$ (left) and two-phase (right) checkerboards and several data λu_0. Dotted: slope ndof^{-1}, i.e. $O(h^2)$. For $p(x)$ with $\lambda = 1$ the width is at round-off level.

$O(\text{ndof}^{-1})$ reported in [8, Exp. 1]); for $\lambda = 0.75$, the ratios $(3.0 \to 3.84)$ are still pre-asymptotic. For the two-phase problem with $\lambda \in \{0.1, 0.2\}$, however, the ratios are only 3.2–3.5 (slope ≈ -0.85), also on $n = 512$, and both bounds close at the same reduced rate. We expected 4 and have no explanation. The effect is not caused by the solver: the final discrete duality gaps of these runs are below 3×10^{-12}, four orders of magnitude below the widths, which are themselves below 2.5×10^{-8}. The rate of closure alone, therefore, does not separate gap from no gap; in all our runs its trend under refinement did (an observation, not a theorem).

Near the thresholds, a second surprise. The computations of [8] indicate a gap for $p(x)$ with $\lambda > 1$ and for the two-phase problem with $\lambda \geq 0.4$. Our widths for $\lambda \in (1, 1.08]$ ($p(x)$) and $\lambda \in \{0.4, 0.5, 0.6\}$ (two-phase), however, still decrease steadily on uniform meshes up to $n = 256$, with ratios 2.0–2.5 that decline only slowly, so they neither confirm nor exclude a gap there. Rigorously, the gap is at most 1.3×10^{-5} for $p(x)$ with $\lambda = 1.08$, and at most 1.2×10^{-4} for the two-phase problem with $\lambda = 0.6$.

Figure 6 compares our widths with the differences $F(u_c) - F_h(u_{nc})$ of [8, Fig. 1], which we read from the vector graphics, identifying the color of each curve from the legend samples. Their adaptive meshes reach values 10–100 times smaller than our uniform meshes. Since their initial meshes resolve the exponent, their conforming energies are exact and their CR energies have $f = 0$, Proposition 5.1 and Theorem D make their differences upper bounds for the gap as well, provided their discrete CR problems are solved accurately. Read in this way, their data bound the gap at $\lambda = 1.08$ by about 4×10^{-7} (observed). Neither data set shows a plateau for $\lambda \leq 1.08$. In both, however, the closure rate *decreases* under refinement for $\lambda \geq 1.04$ (our ratios $2.28 \to 2.21$, $2.24 \to 2.11$, $2.21 \to 2.02$; their slope for $\lambda = 1.08$ drops from about -1.1 to -0.45), whereas it is steady or increasing for $\lambda = 0.99$; this is the pattern of the gap cases. Since $m_{\mathcal{H}} - m_{\mathcal{W}}$ vanishes at $\lambda = 1$, a gap below the final widths is compatible with all the data. We conclude only that the gap for $\lambda \in (1, 1.08]$, if present, is below 1.3×10^{-5} (our bound) and plausibly below 4×10^{-7}; it cannot be certified either way. The widths stall, which is the signature of a gap but not a proof of one (Remark 5.6(c)), for $\lambda \geq 1.5$ ($p(x)$) and $\lambda \geq 1$ (two-phase).

Quadrature. The guarantees depend on how the coefficients are sampled. Reproducing [8, §4.2] ($p(x)$, $\lambda = 5$), we find that replacing the exponent on the nodal patch of the origin by its maximum

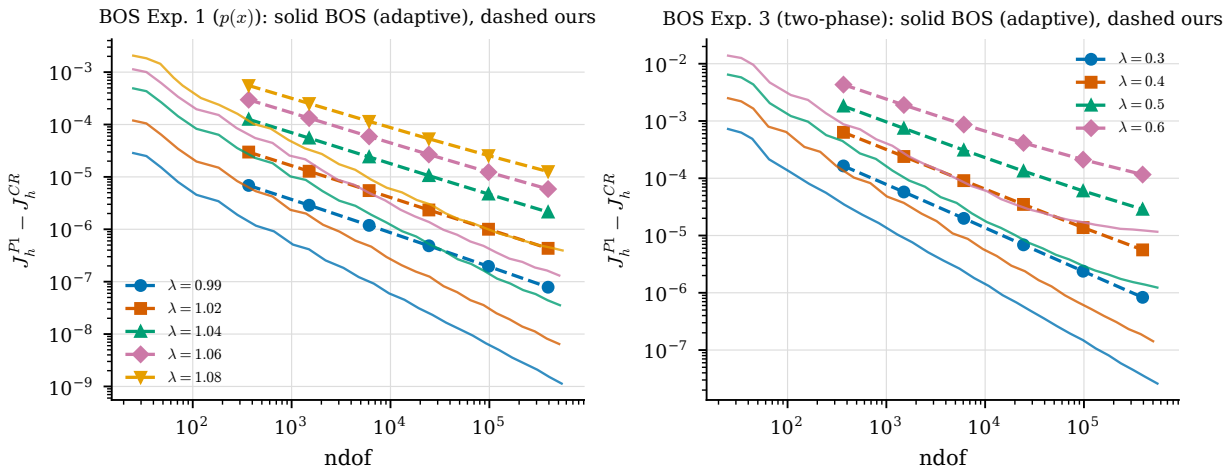


FIGURE 6. Near the gap thresholds reported in [8]: our certified widths $J_h^{\mathrm{P1}} - D(\sigma_n)$ (uniform meshes; dashed, markers) and the differences digitized from [8, Fig. 1] (adaptive meshes; solid) versus the number of unknowns. Left: $p(x)$ checkerboard, $\lambda = 0.99, \dots, 1.08$; right: two-phase checkerboard, $\lambda = 0.3, \dots, 0.6$. Our widths equal $J_h^{\mathrm{P1}} - J_h^{\mathrm{CR}}$ to within 3×10^{-12}.

sends *both* P1 and CR towards the P1 value (59.73 for exact sampling on $n = 256$; perturbed: P1 60.66, CR 59.91), and replacing it by its minimum sends both towards the CR value (45.93; perturbed: 45.85, 45.52). The perturbed energies are not bounds for the original problem: the guarantees of Proposition 5.1 and Theorem D require a majorant sampling for P1 and a minorant sampling for CR (Remark 5.6(d)).

6.6. **The size of the discrete saddle points.** The second diagnostic is the size of the discrete saddle points. Figure 7 shows the quantities of Theorem 5.7. In the gap regime, $\max|\eta_h^{\mathrm{P1}}|$ grows like $h^{-0.9}$ (e.g. $104.8 \to 367.4 \to 706.4$ for $p(x)$, $\lambda = 3$, $n = 64, 256, 512$, and $138 \to 492 \to 951$ for the two-phase problem), consistently with Theorem 5.7(d); recall that the gap for these data is inferred, not proved. The CR flux grows much more slowly ($11.9 \to 22.7 \to 31.6$). A flux that grows like $h^{-0.9}$ at a single point is consistent with a singularity $|\eta| \sim |x|^{-0.9}$, which lies in L^2, so this growth says nothing against (S). Growth of $\max|\eta_h|$ alone is not a gap indicator either: it grows mildly also without a gap ($0.54 \to 0.57 \to 0.59$ for the two-phase problem with $\lambda = 0.2$). The observed rates differ by an order of magnitude, but we have no theory for them.

The L^2 norms behave differently. The CR gradient norm $\|\nabla_h u_h^{\mathrm{CR}}\|$ increases slowly ($7.57 \to 7.73 \to 7.82$), roughly like $(\log(1/h))^{1/2}$, which is consistent with, but does not establish, an unbounded CR family. The P1 norms $\|\nabla u_h\|$ ($6.565 \to 6.612 \to 6.627$) and $\|\eta_h\|$ ($9.145 \to 9.158 \to 9.212$) look bounded up to $n = 512$, except for the last increment of $\|\eta_h\|$. By Theorem 5.7(a), bounded P1 saddle points would imply (S) together with $m_{\mathcal{H}} = m_{H^1}$; if the gap inferred above is real, this would be an instance of (S) *with* a gap, in line with the discussion after Proposition 3.12. The data are too short to decide.

6.7. **Beyond the checkerboards.** The checkerboards have piecewise constant coefficients and polynomial growth. Two further examples test what happens when either property is lost.

Double phase with the Esposito–Leonetti–Mingione weight. Let $G = t^{1.5}/1.5 + a(x)t^q/q$ with $a = \max(x_2^2 - x_1^2, 0)/|x|$, which is Lipschitz (so $\alpha = 1$), data λu_0 and $f = 0$. The no-gap range of [20, Thm. 2.3] is then $q \le 2.5$, while $q = 4 > d + \alpha$ lies in the range of the counterexamples of [39]. Since a is not piecewise constant, a rigorous bracket is obtained from a majorant sampling for P1

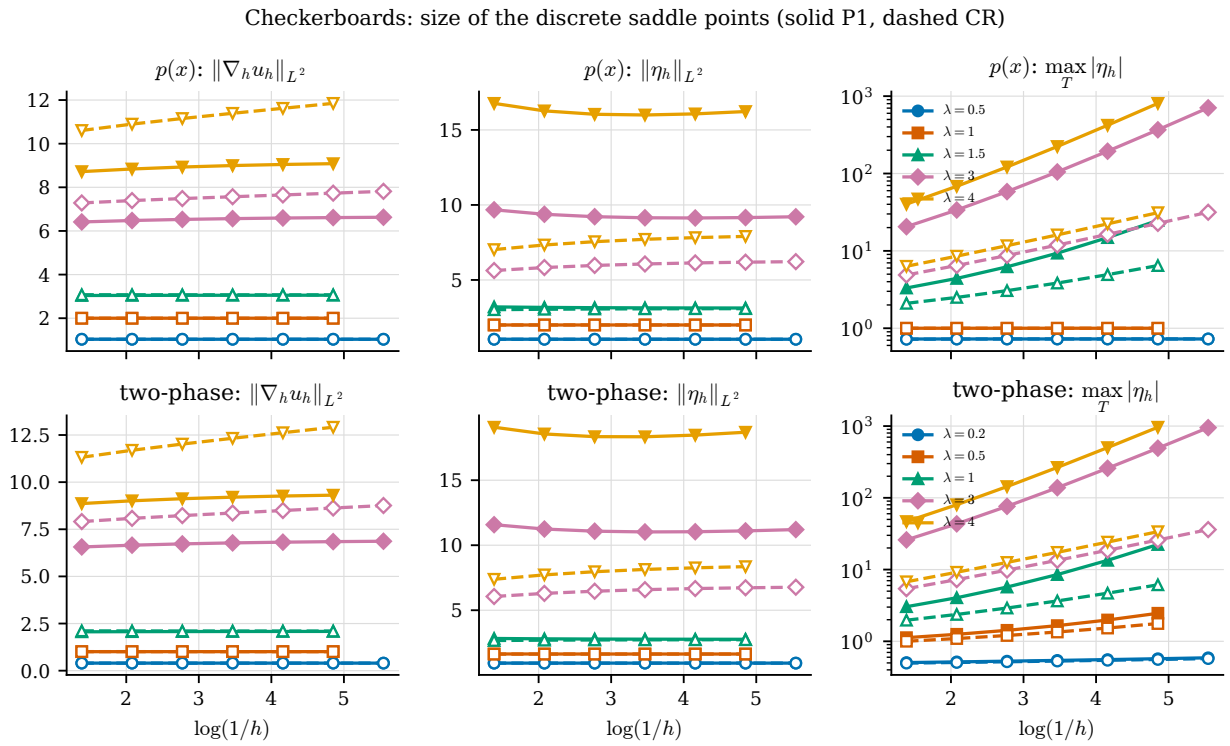


FIGURE 7. Size of the discrete saddle points for the $p(x)$ (top) and two-phase (bottom) checkerboards, P1 (solid) and CR (dashed), several data λu_0, versus $\log(1/h)$: $\|\nabla_h u_h\|_{L^2}$ (left), $\|\eta_h\|_{L^2}$ (center) and $\max_T|\eta_h|$ (right, logarithmic scale).

and a minorant sampling for CR (Remarks 5.2(c) and 5.6(d)). Its width, however, is dominated by the $O(h)$ sampling error: the bracket is, for example, $[82.15, 82.85]$ for $q = 2.4$, $\lambda = 8$, $n = 256$, and $[101.6, 196.6]$ for $q = 4$, $\lambda = 8$ (gap ≤ 95.0). The difference of the barycenter-sampled P1 and CR energies, for which we have no theory, is more informative. For $q = 2.4$ it closes at $O(h^2)$ (ratio $\to 4$) for all $\lambda \leq 8$. For $q = 4$ it stalls for $\lambda \geq 4$ (ratios $1.59 \to 1.30$ for $\lambda = 4$), while $\max|\eta_h^{\mathrm{P1}}|$ grows; it is ambiguous at $\lambda = 2$ (steady ratio ≈ 2.45) and closes at $O(h^2)$ for $\lambda \leq 1$. For $q = 2.8$ and $\lambda = 8$ it closes at a ratio 2.4–2.6 that increases with n, so the case remains undecided for $n \leq 128$.

An exponential checkerboard. The integrand $G = a(x)(e^t - 1 - t) + t^2/2$ with $a = \mathbf{1}_{\{|x_1|<|x_2|\}}$ satisfies neither the growth condition of [8] nor Δ_2, but Theorem D still applies. At $n = 256$ the bracket is $[0.740687, 0.740812]$ for $\lambda = 0.5$ and $[35.18, 37.45]$ for $\lambda = 3$, and the width stalls for $\lambda \geq 2$ (ratio 1.13 at $\lambda = 3$), the pattern of the gap cases above. We know of no theory for the gap in this problem.

6.8. **Discussion.** The experiments confirm the division of labor suggested by the theory. The solver is never the problem: the iteration converged in every Lavrentiev run, for every r, as Theorem C guarantees, and a failure of (S) showed up only as iteration counts growing under refinement (observed for one example, Section 6.3). What a gap changes is the meaning of the computed numbers. Here the bracket of Corollary 5.5 turns two schemes that converge separately, but to different targets, into a computable statement about $m_{\mathcal{W}}$ and $m_{\mathcal{H}}$. Three observations did not match our expectations, and we report them as such. The no-gap two-phase controls close at a ratio 3.2–3.5 rather than 4; near the reported thresholds, neither data set shows a plateau, although both show a decreasing closure rate; and the P1 saddle points in the gap regime look bounded in L^2 up to $n = 512$, which, if both this boundedness and the gap are real, would be an instance of

(S) with a gap. Finally, we recall the limitations of the experiments: widths certify upper bounds only, extrapolations are heuristic, and the disk errors are computed on inscribed polygons against radial solutions computed by quadrature.

7. Conclusions and open problems

For an arbitrary convex Young integrand and $\rho = r$, the convergence of Glowinski's method at the continuous level is equivalent to the boundedness of the iterates, and it hinges on the regularity assumption (S). Under (S), the iteration converges weakly in general and strongly under Lieberman's condition, without any continuity in x. Assumption (S), however, is not available in general for double phase, two-phase and exponential energies, and it is a different property from the absence of a Lavrentiev gap. At the discrete level, the iteration converges unconditionally, for P1 and CR alike, and the remaining issue is consistency, which is governed by the gap. Under the assumptions of [8, Thm. 1] (polynomial growth, resolved or minimum quadrature), CR elements give the method a convergence theory in the Lavrentiev regime (Theorem C and [8, Thm. 1]). Moreover, the CR iterates carry a guaranteed lower bound for $m_{\mathcal{W}}$, which, together with the P1 energy, brackets the gap.

Several problems remain open. (1) The behavior of u_n when (S) fails, in which case (η_n, ν_n) is unbounded; cf. [46, Rem. 23]. (2) Strong convergence of u_n for general convex G, the case $\rho \neq r$ without Glowinski's hypotheses (iii)–(iv), and rates for the iteration, including their dependence on h and r and a theory for the choice of r. (3) Whether (S) can hold together with a gap for loads $f \in L^d$; Example 3.8 is degenerate, and the P1 data of Section 6.6 are suggestive but not conclusive. (4) A computable *lower* bound for $m_{\mathcal{H}}$, which would certify the positivity of a gap. (5) Rates for the width Δ_h, an explanation of the observed $O(h^2)$ closure without a gap and of its reduced rate for the two-phase controls, and an extension of Theorem D to $f \neq 0$ for the CR energy itself and to non-resolved coefficients with sampling errors smaller than $O(h)$. (6) Global regularity, and hence (S), for Musielak–Orlicz integrands $G(x,t)$ without Δ_2 and for double phase energies up to the boundary.

Software availability. The Python package `glaplace` (NumPy/SciPy; MIT license) implements the method with P1 and CR elements for arbitrary Young functions on two-dimensional triangulations, including the certified lower bound of Theorem D. It contains the scripts and raw data of all experiments of Section 6 and is available at `https://github.com/jfbonder/glaplace`. It generalizes the code for the $p(x)$-Laplacian of [4].

Declaration of the use of AI tools. Generative AI was used extensively in this work, in the form of Claude models developed by Anthropic, accessed through the Claude desktop application and its agent tools (October 2026). The research problem, the choice of the method and of the test problems, and the scientific direction are the authors'. The AI tools were used for the following tasks: (i) writing the Python code of the solver and of all numerical experiments, including the tests, starting from the authors' original MATLAB code for the $p(x)$-Laplacian (the code of [4]), which was translated to Python and extended to general Young functions and to Crouzeix–Raviart elements; (ii) drafting the proofs of Theorems A–D and of the auxiliary results, starting from the authors' earlier unpublished argument at the continuous level; (iii) searching and compiling the literature, extracting the hypotheses of the cited results from the sources, and checking the bibliographic data; (iv) drafting and editing the text of the article; (v) independent critical reviews of the mathematics and of the exposition, carried out by separate instances of the model; and (vi) in the final rewriting, a team of AI agents with separate roles (numerical analyst, expert in nonlinear partial differential equations, programmer, bibliography reviewer, editor and translator), whose drafts, computations and reports were checked against each other, followed by two further independent referee reports by AI agents (numerical analysis and PDE), whose recommendations

were incorporated into the text. The numerical results reported here were produced by running the code described above. The authors take full responsibility for the content of the article.

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